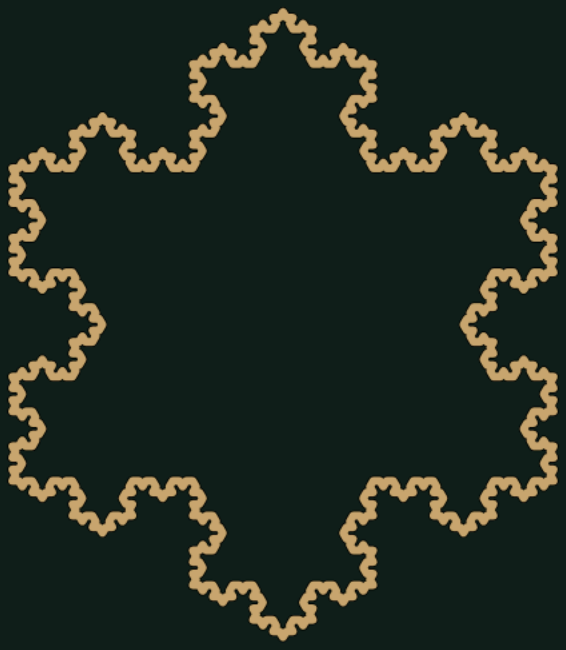
FRACTAL FLORA



A Coloring Book of Algorithmic Growth

50 space-filling and fractal plant designs

MathRelaxing

FRACTAL FLORA

A Coloring Book of Algorithmic Growth

50 L-System Space-Filling Curves & Plants · MathRelaxing

About this book

Every design in these pages was generated using Lindenmayer systems (L-systems) — a mathematical language originally developed to model the growth processes of plant development.

By repeatedly applying a simple set of rewriting rules to a sequence of symbols, complex branching structures and space-filling curves emerge. The patterns grow iteratively, expanding into intricate fractals like the Hilbert Curve, the Dragon Curve, and algorithmic ferns that mimic nature perfectly.

The next page explains the mathematics behind the drawings. After that, 50 unique algorithmic gardens await your colour.

How to colour:

- Any pens, pencils or markers work on the matte paper.
- Follow the continuous lines of the space-filling curves, or colour the enclosed spaces they create.
- Slip a spare sheet behind the page if you use wet markers.

Enjoy the infinite growth.

The mathematics

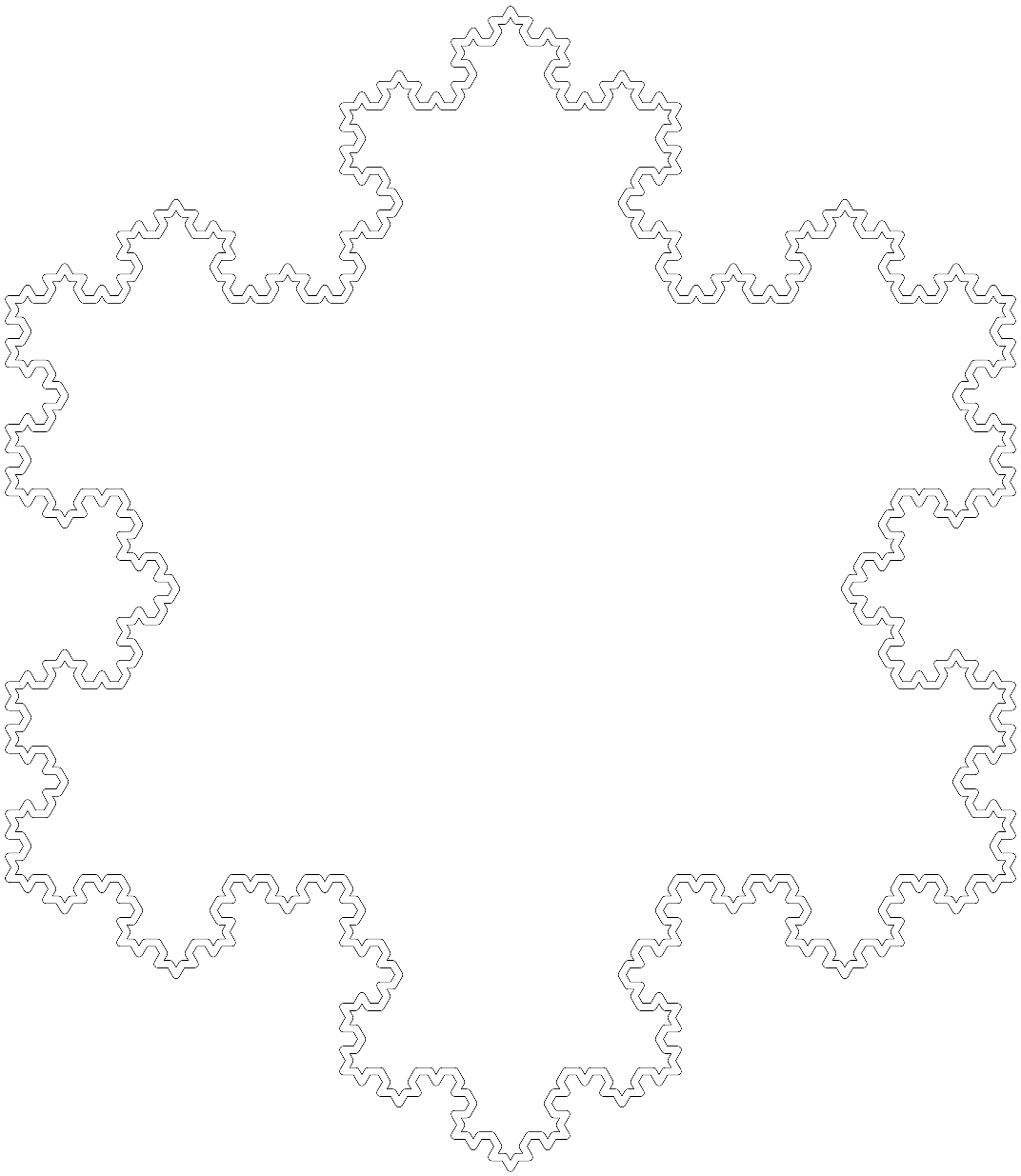
L-Systems. An L-system consists of an alphabet of symbols, an initial string called the axiom, and a set of production rules that expand each symbol into a larger string in parallel.

For example, a simple algae model by Aristid Lindenmayer uses:

Axiom: A
Rules: A → AB, B → A

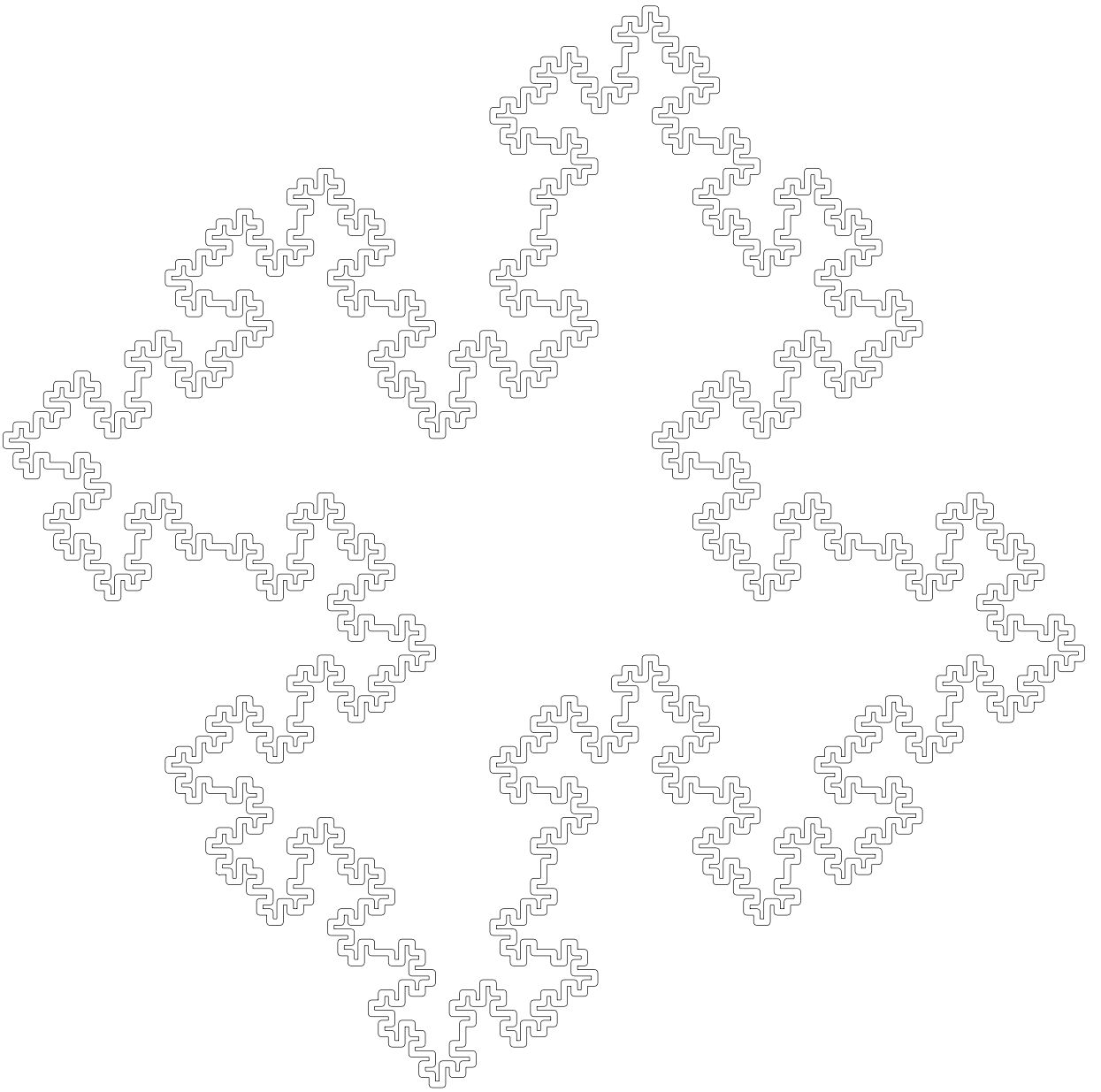
Iteration. Applying the rules repeatedly causes the string to grow exponentially. Iteration 0 is 'A'. Iteration 1 is 'AB'. Iteration 2 is 'ABA'. Iteration 3 is 'ABAAB'.

Turtle Graphics. To draw the fractals, we read the final string as commands for a 'turtle' holding a pen. 'F' means draw forward. '+' means turn right by an angle. '[' means remember current position, and ']' means return to it. This simple geometry interprets the string into the beautiful botanical branching shapes and continuous curves you see in this book.



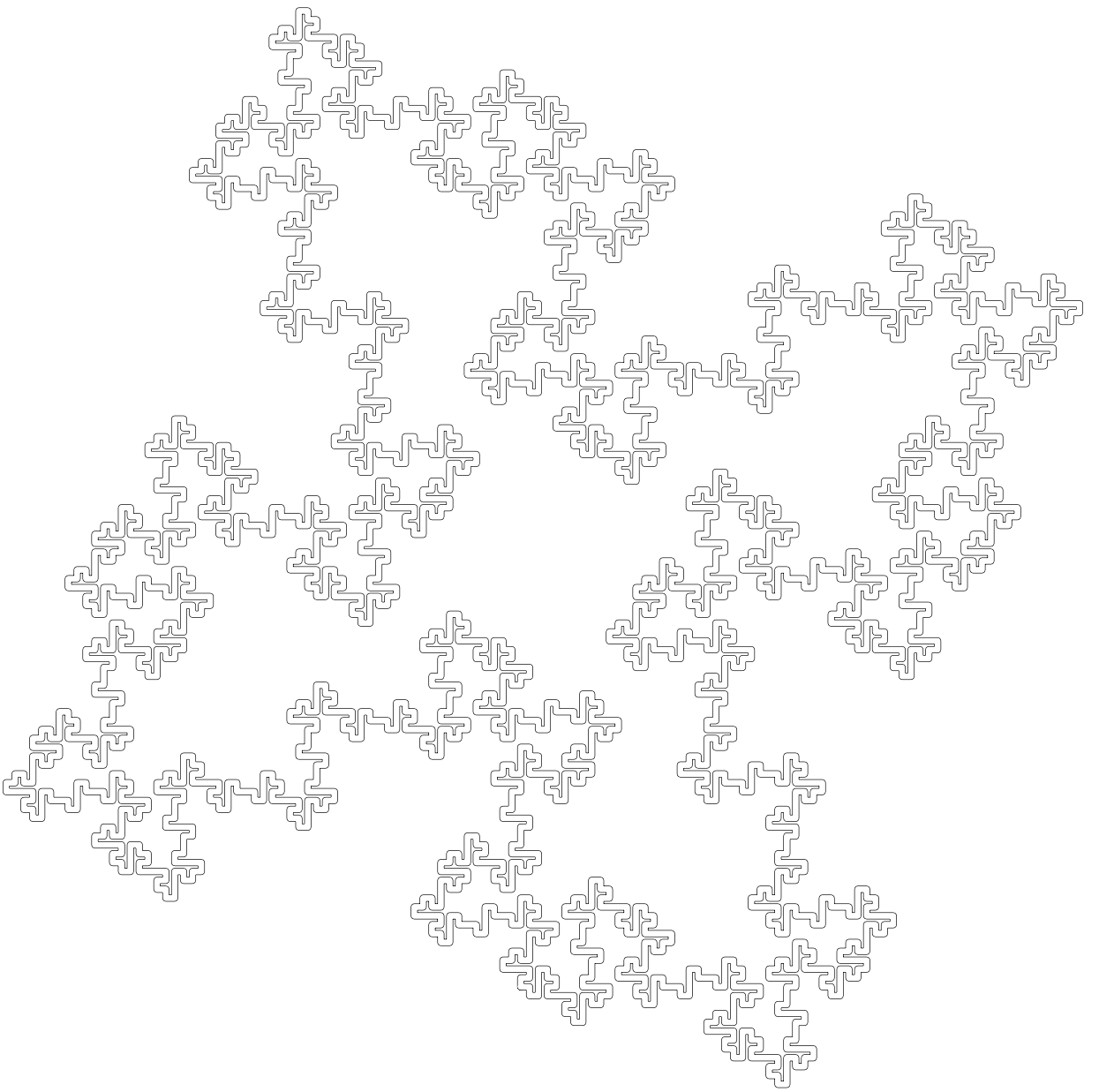
Koch Mandala

L-System Generator • 4 Iterations



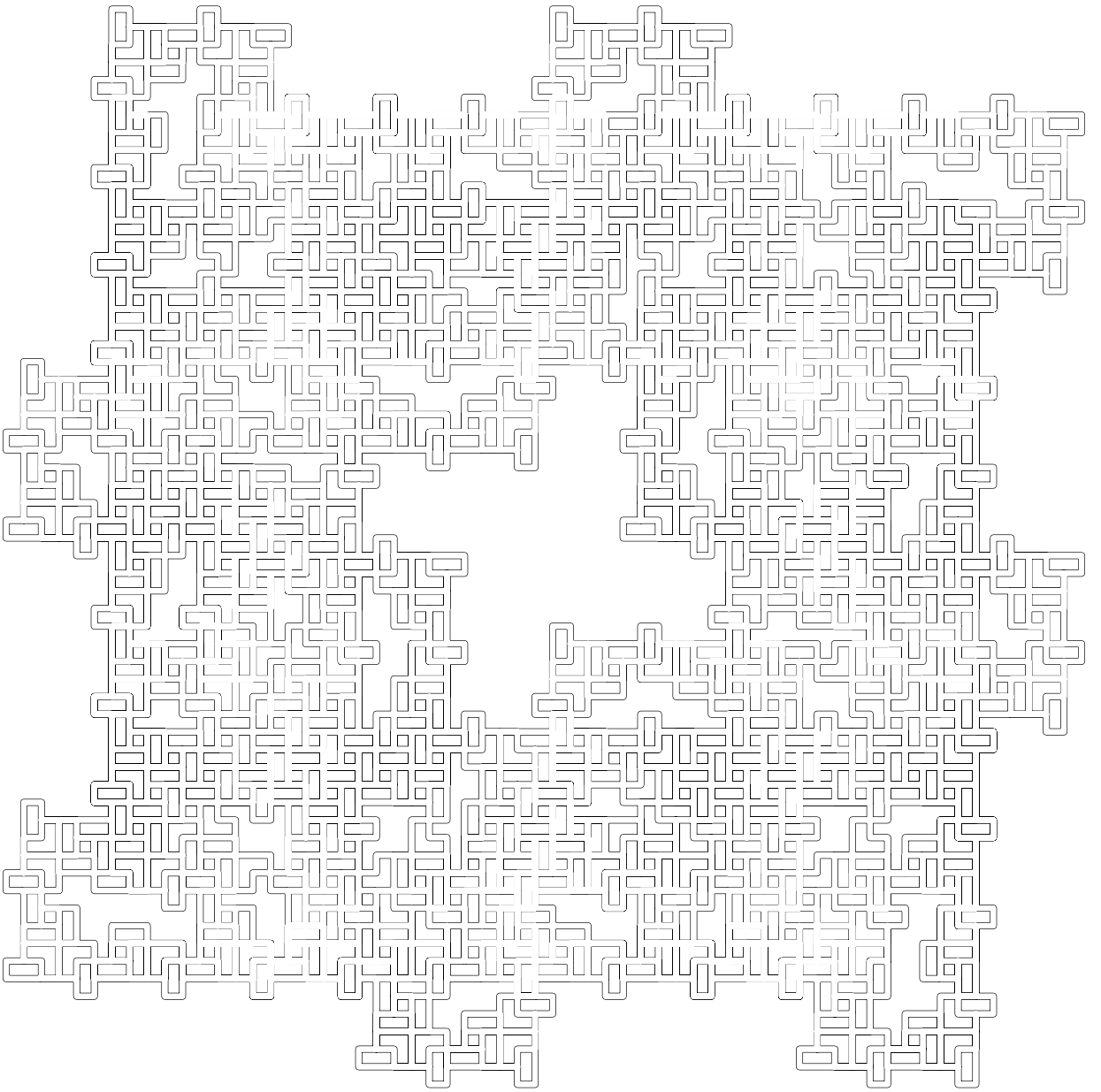
Minkowski Loop

L-System Generator • 3 Iterations



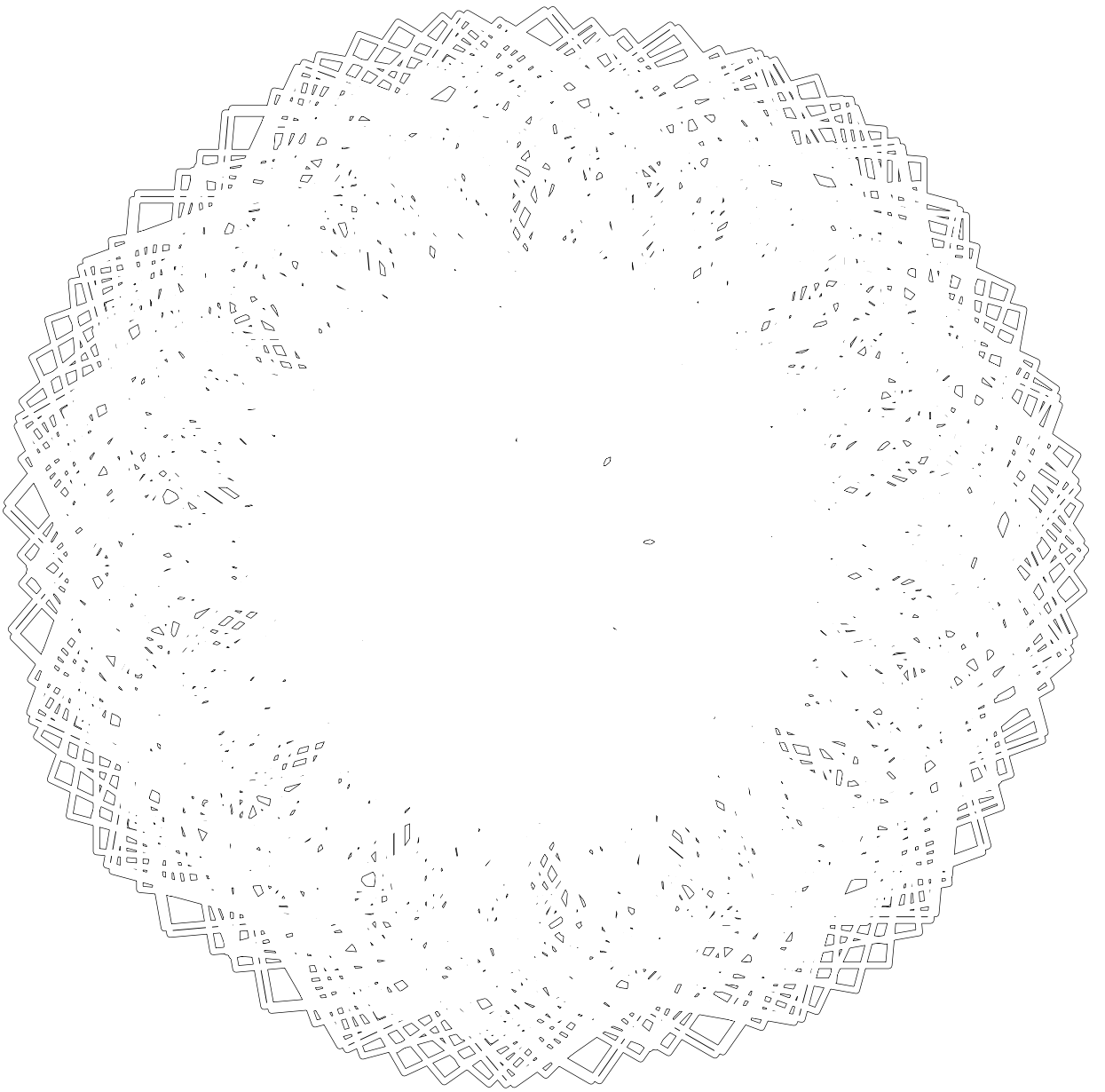
Quadratic Koch Island

L-System Generator • 3 Iterations



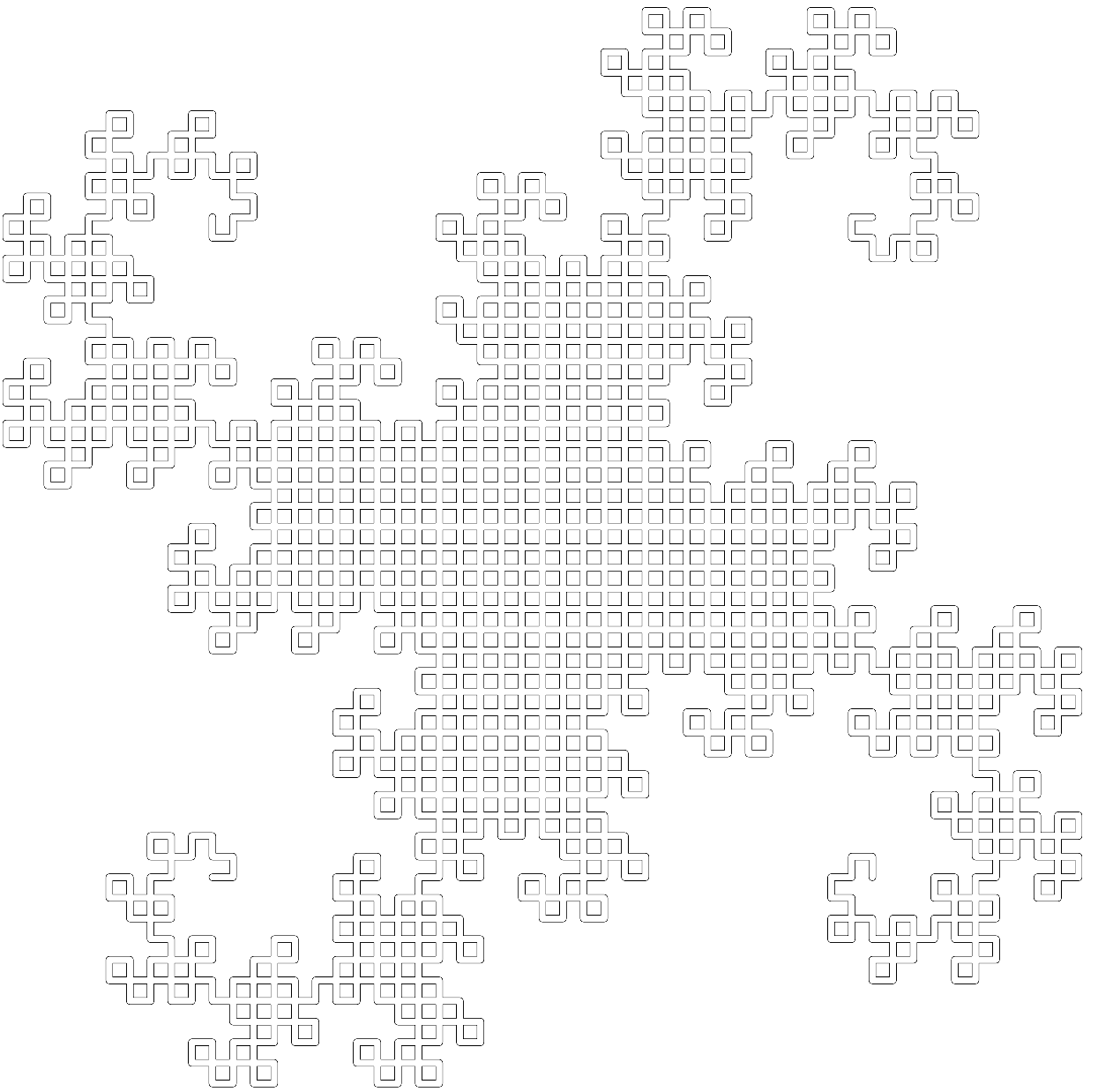
Koch Square

L-System Generator • 4 Iterations



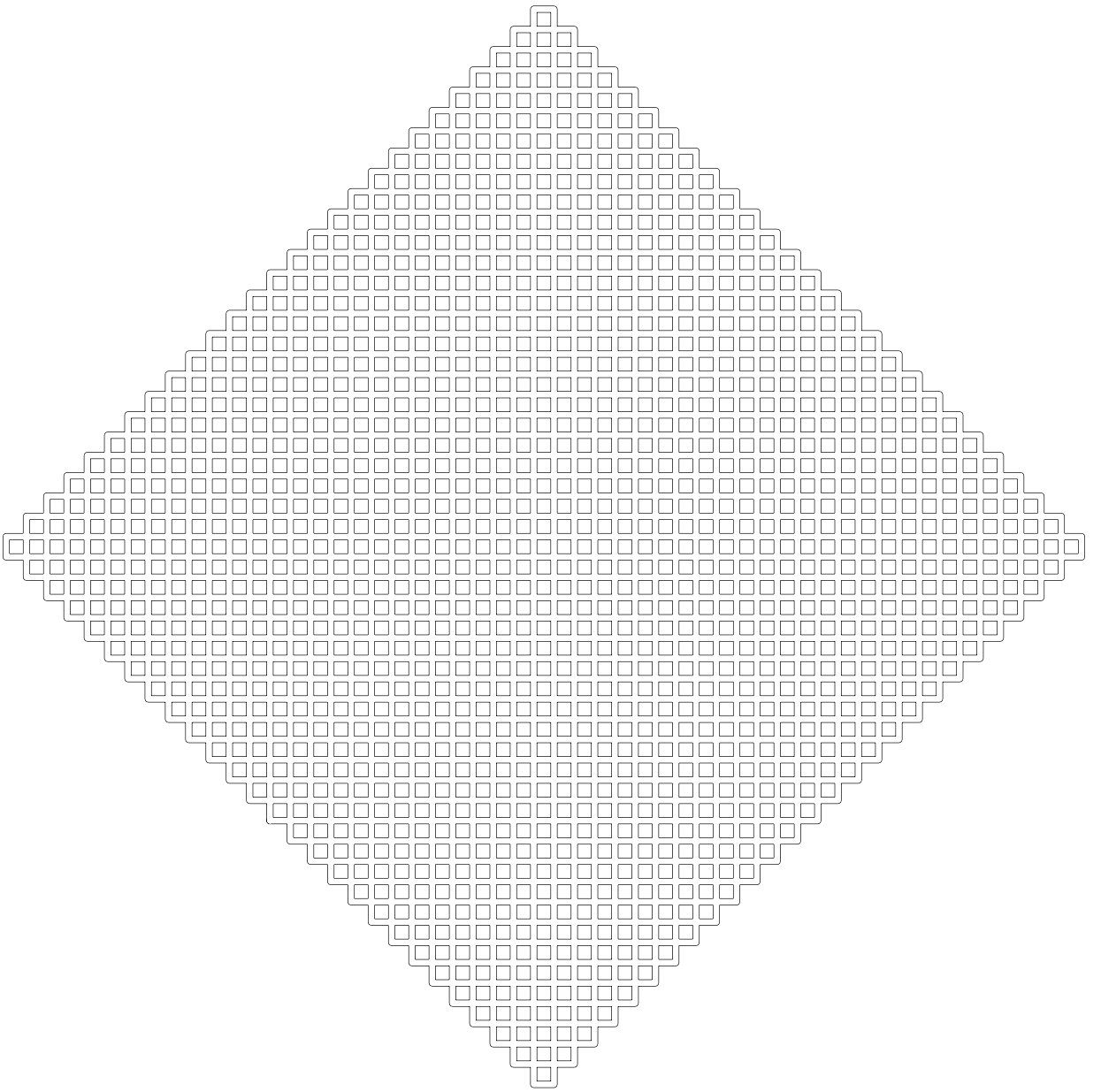
Cesaro Fractal

L-System Generator • 5 Iterations



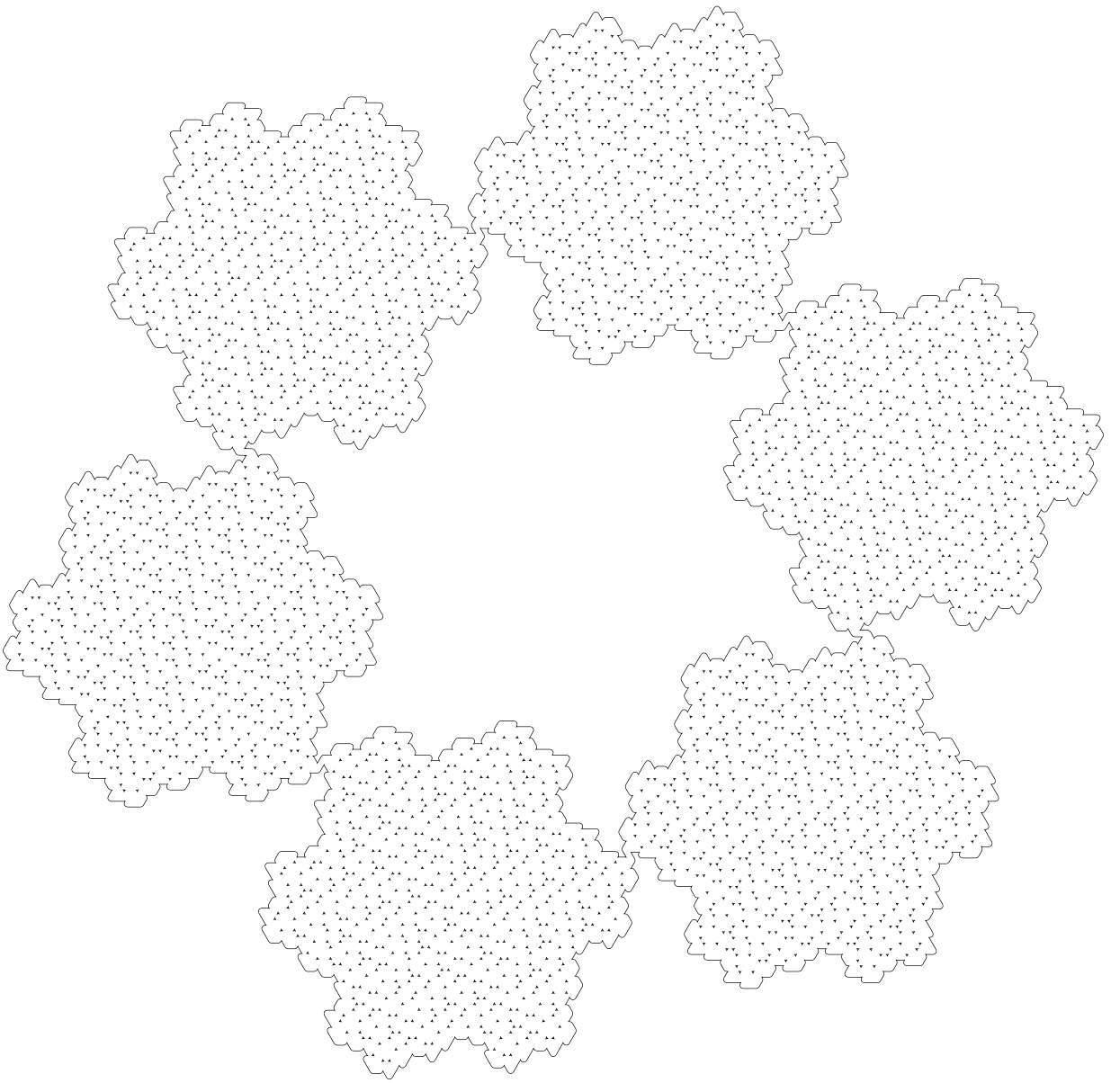
Dragon Loop

L-System Generator • 10 Iterations



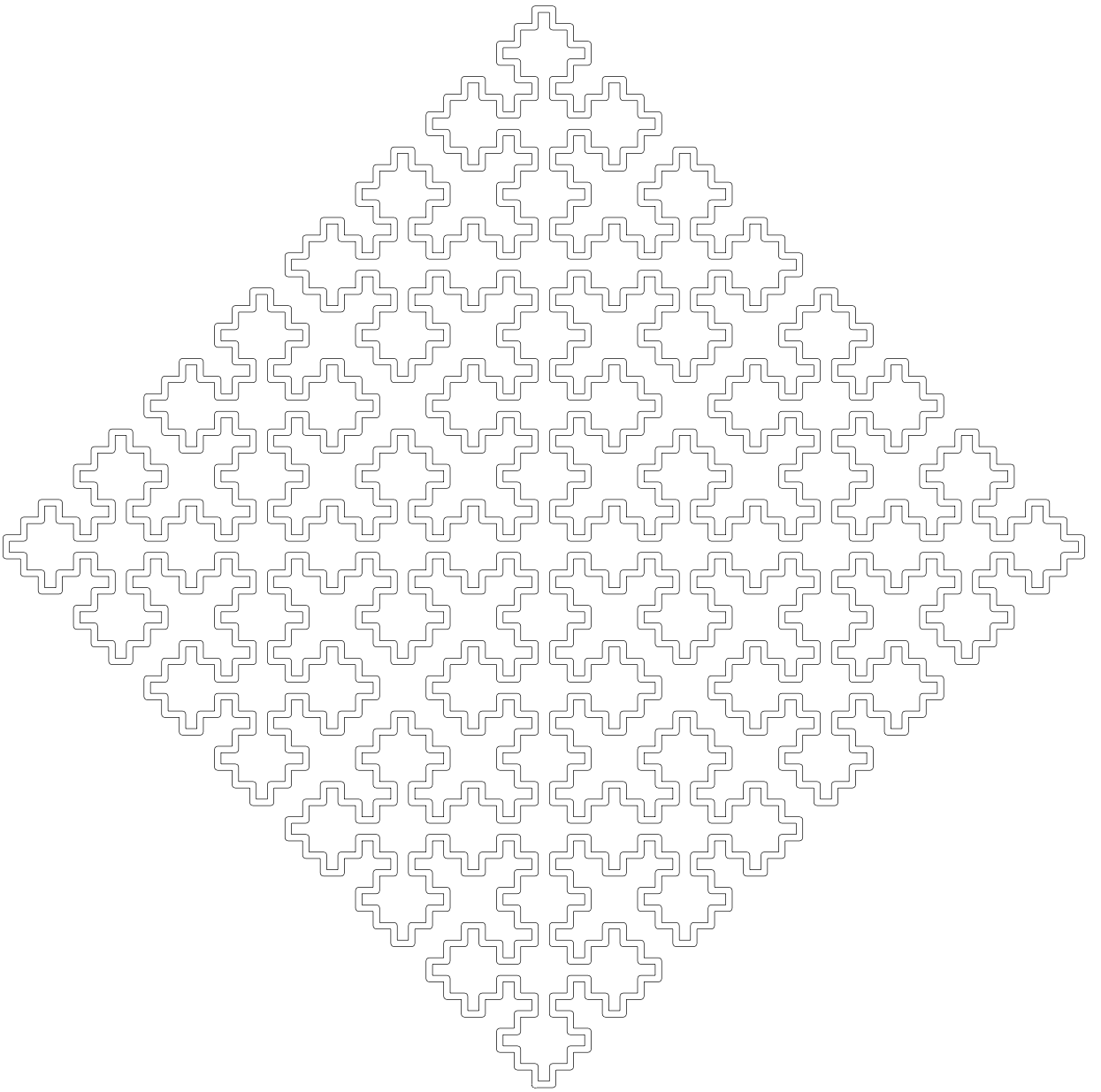
Peano Loop

L-System Generator • 3 Iterations



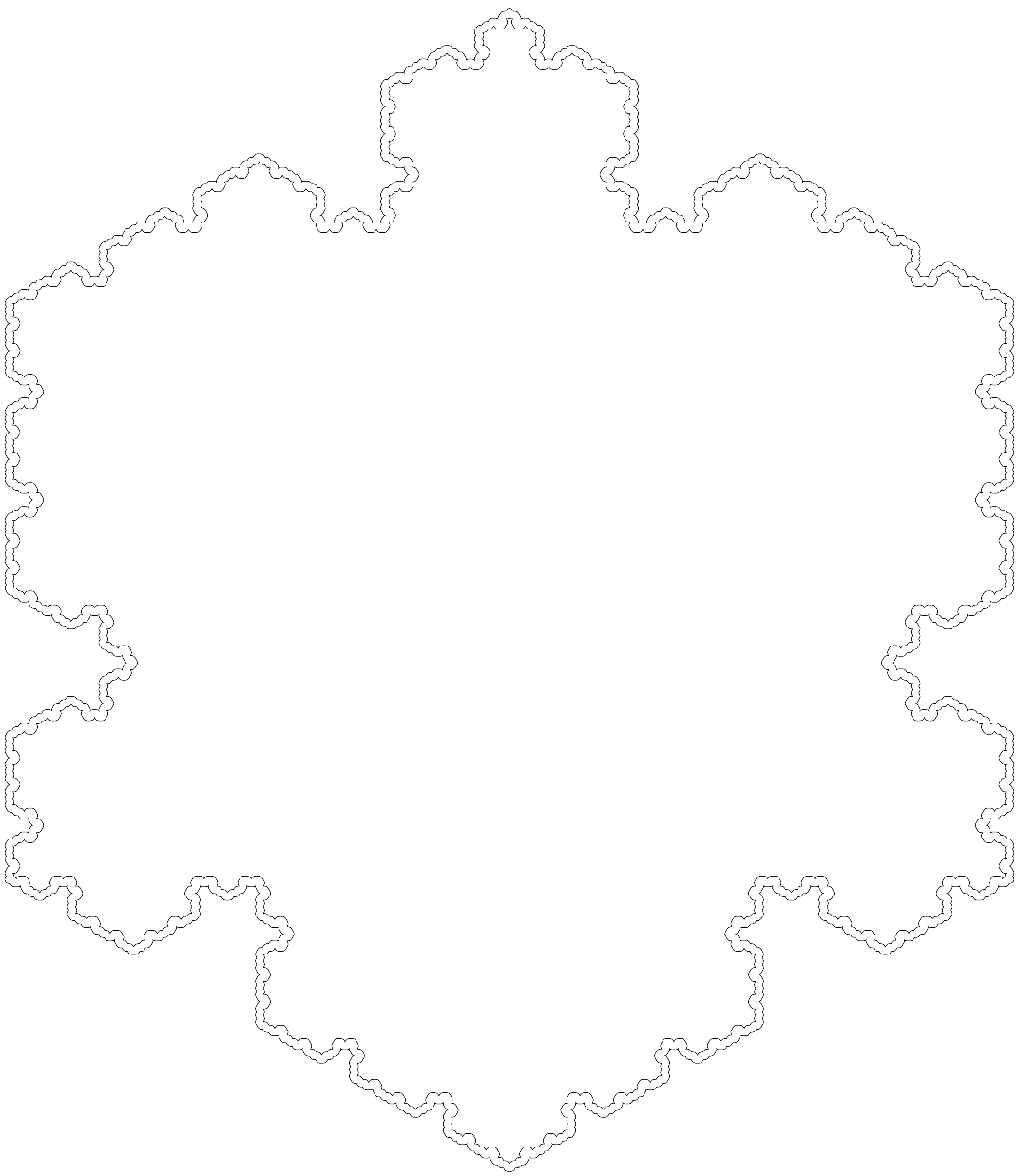
Gosper Ring

L-System Generator • 4 Iterations



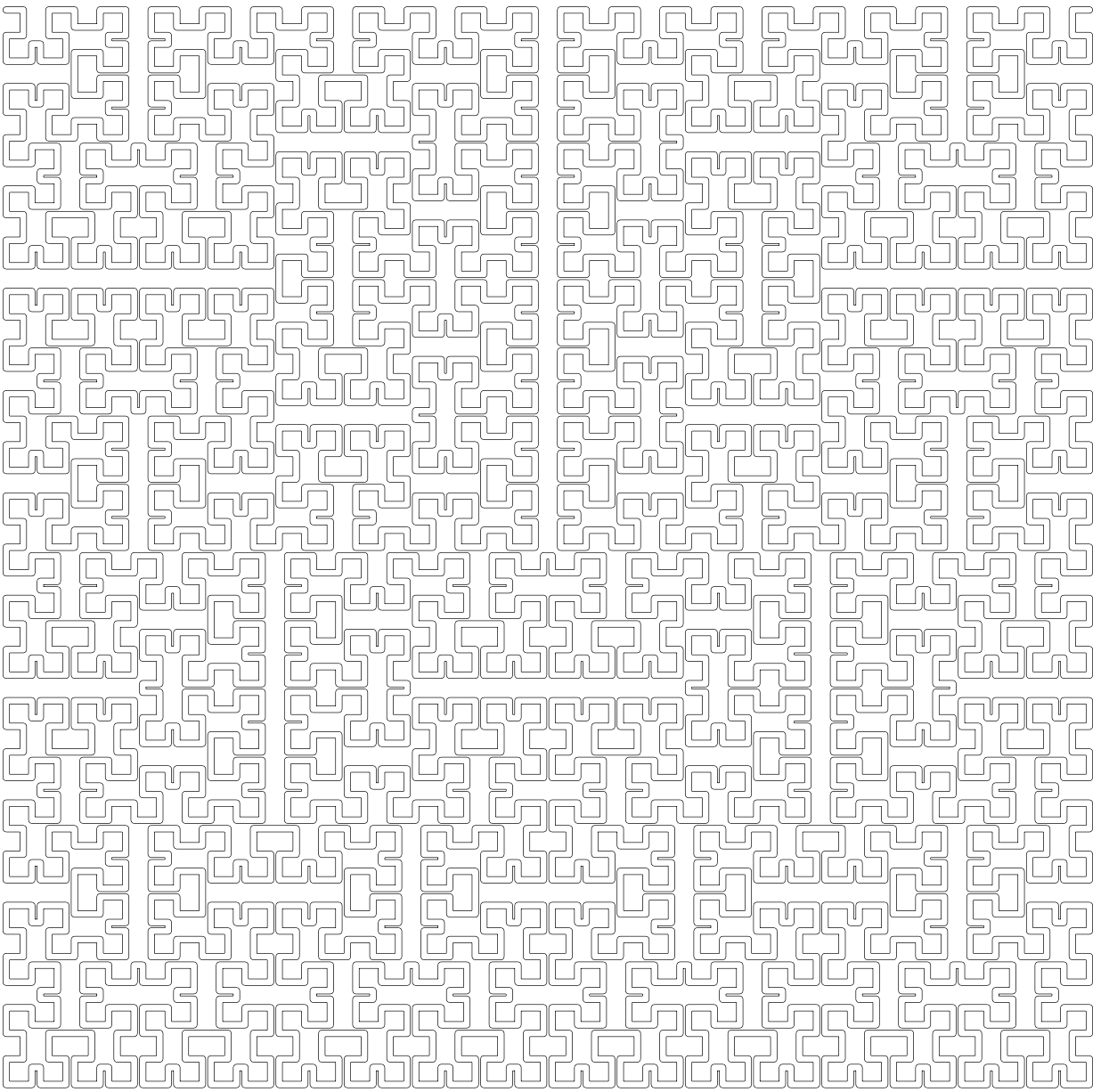
Sierpinski Ring

L-System Generator • 4 Iterations



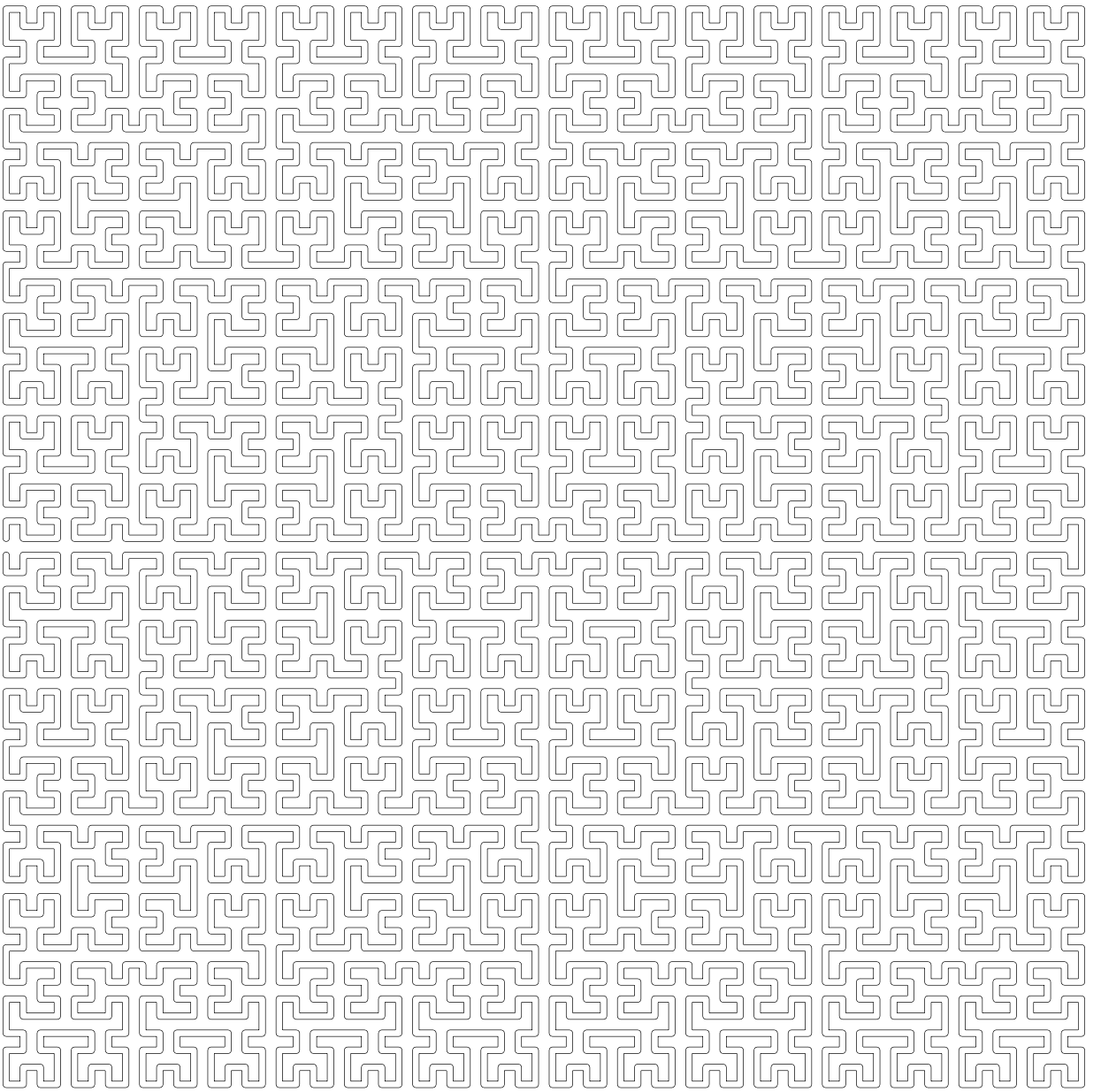
Hexagonal Snowflake

L-System Generator • 4 Iterations



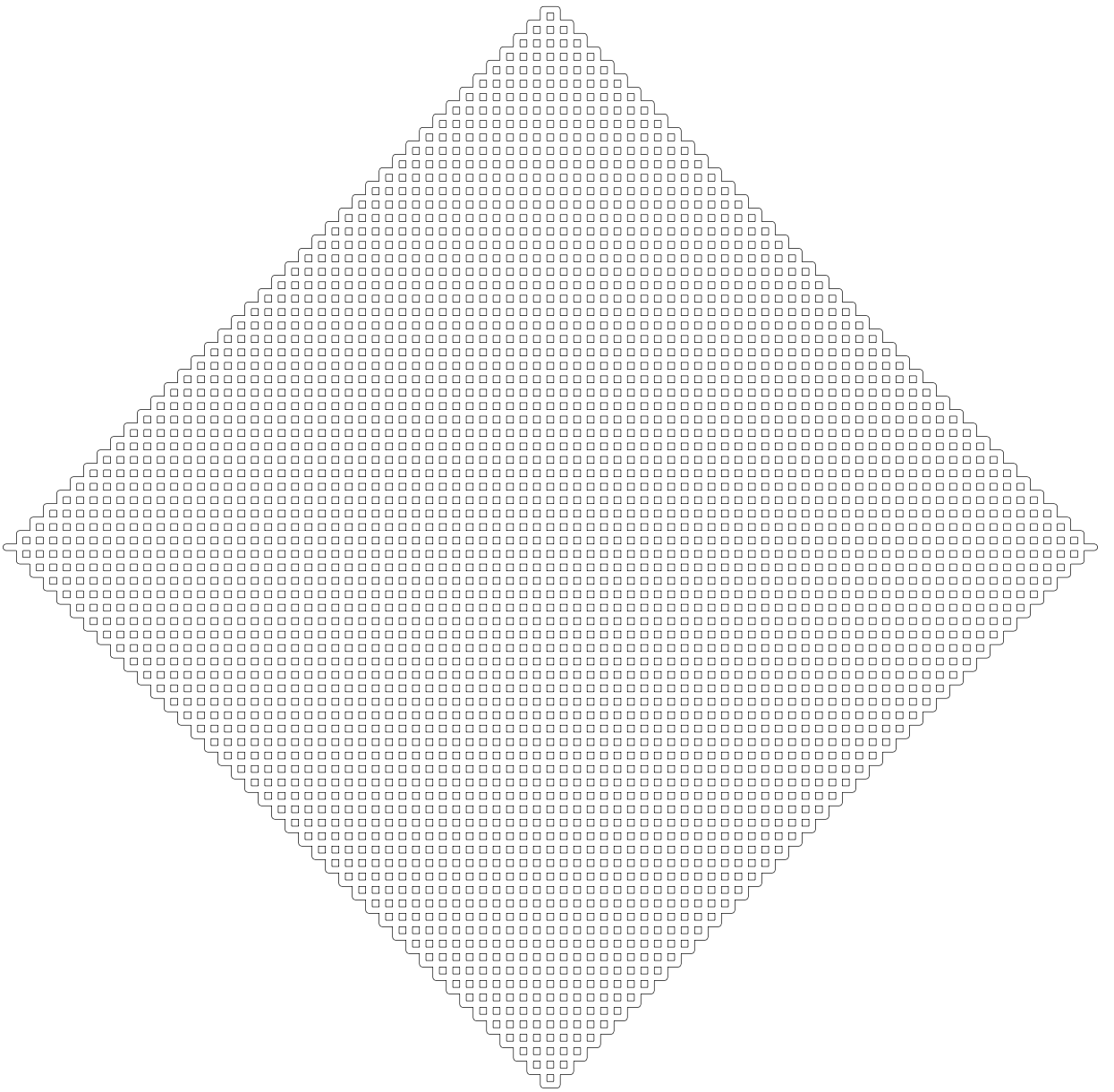
Hilbert Curve

L-System Generator • 6 Iterations



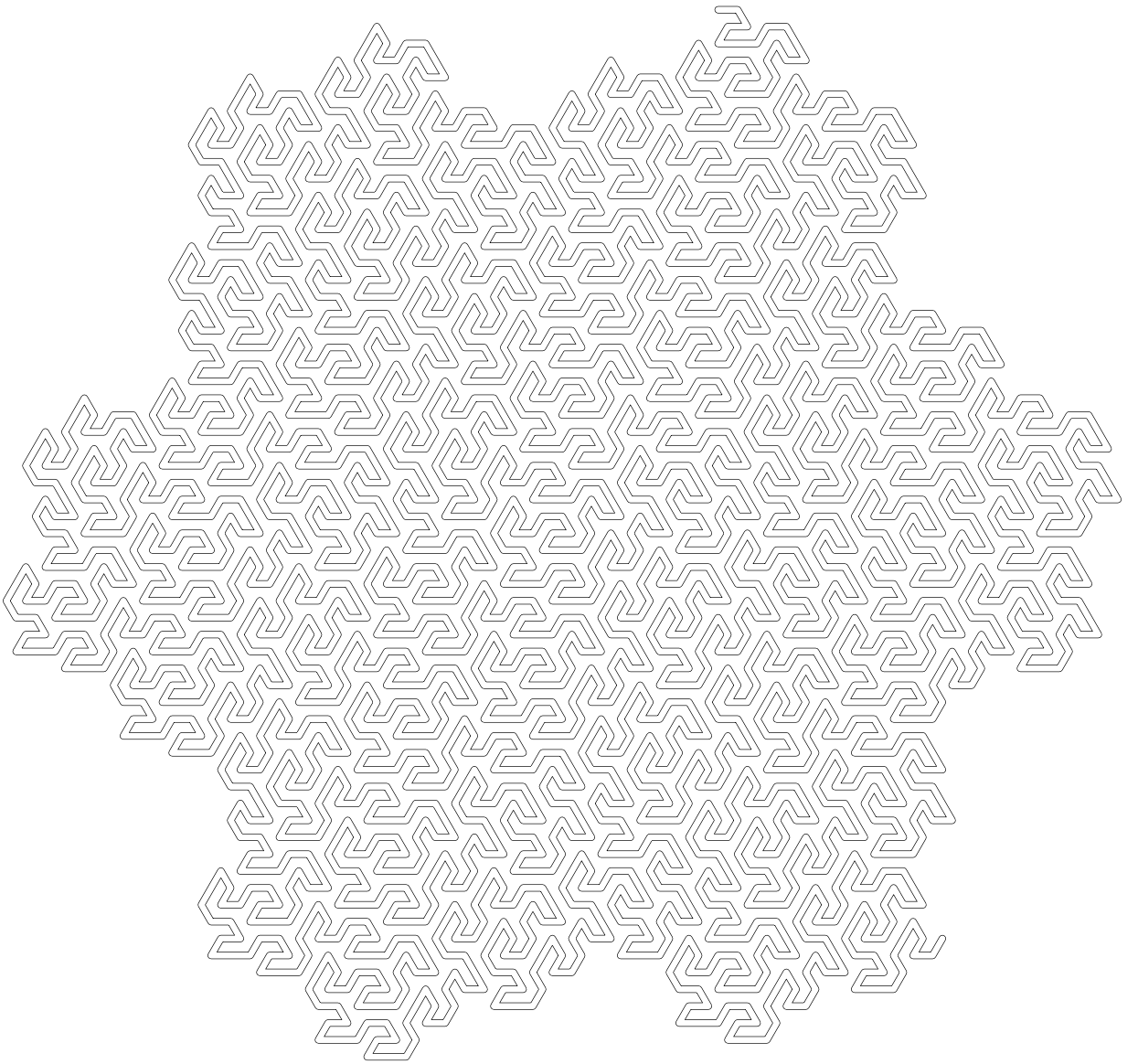
Moore Curve

L-System Generator • 5 Iterations



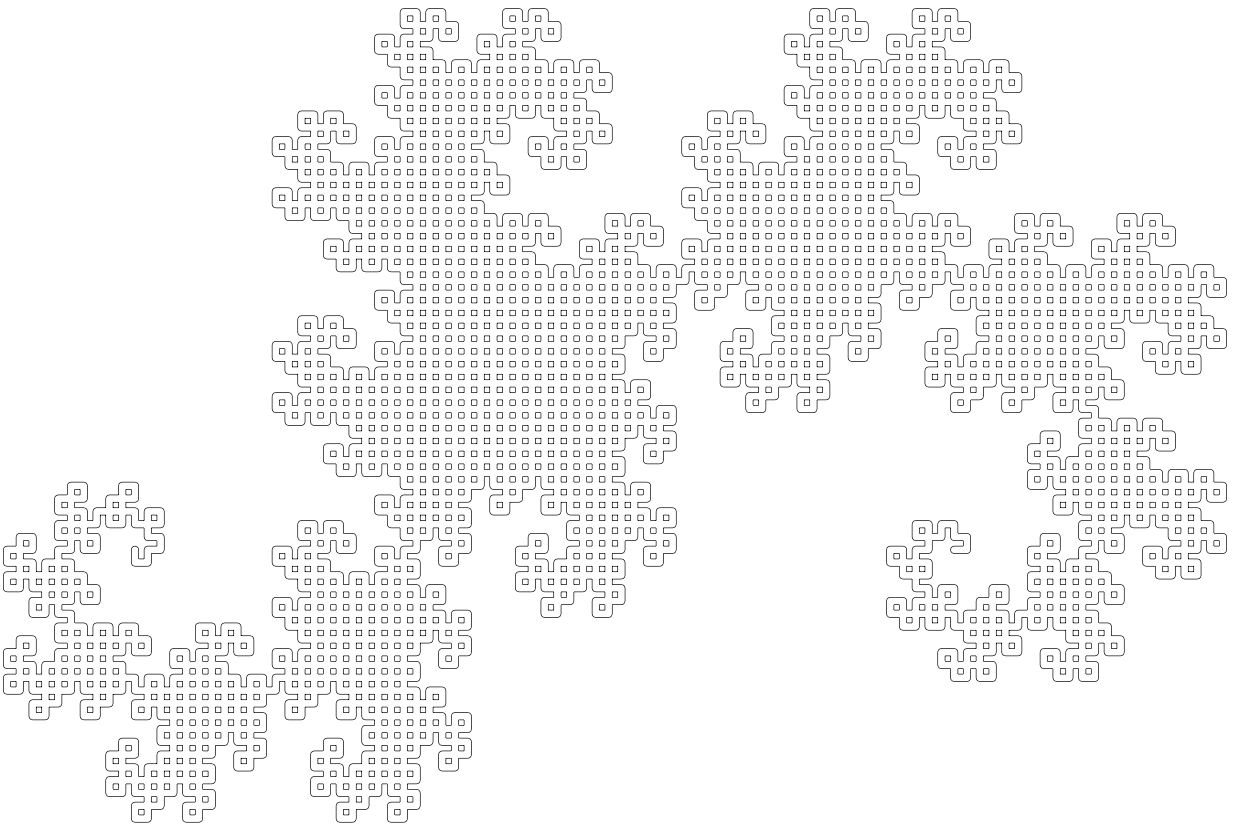
Peano Curve

L-System Generator • 4 Iterations



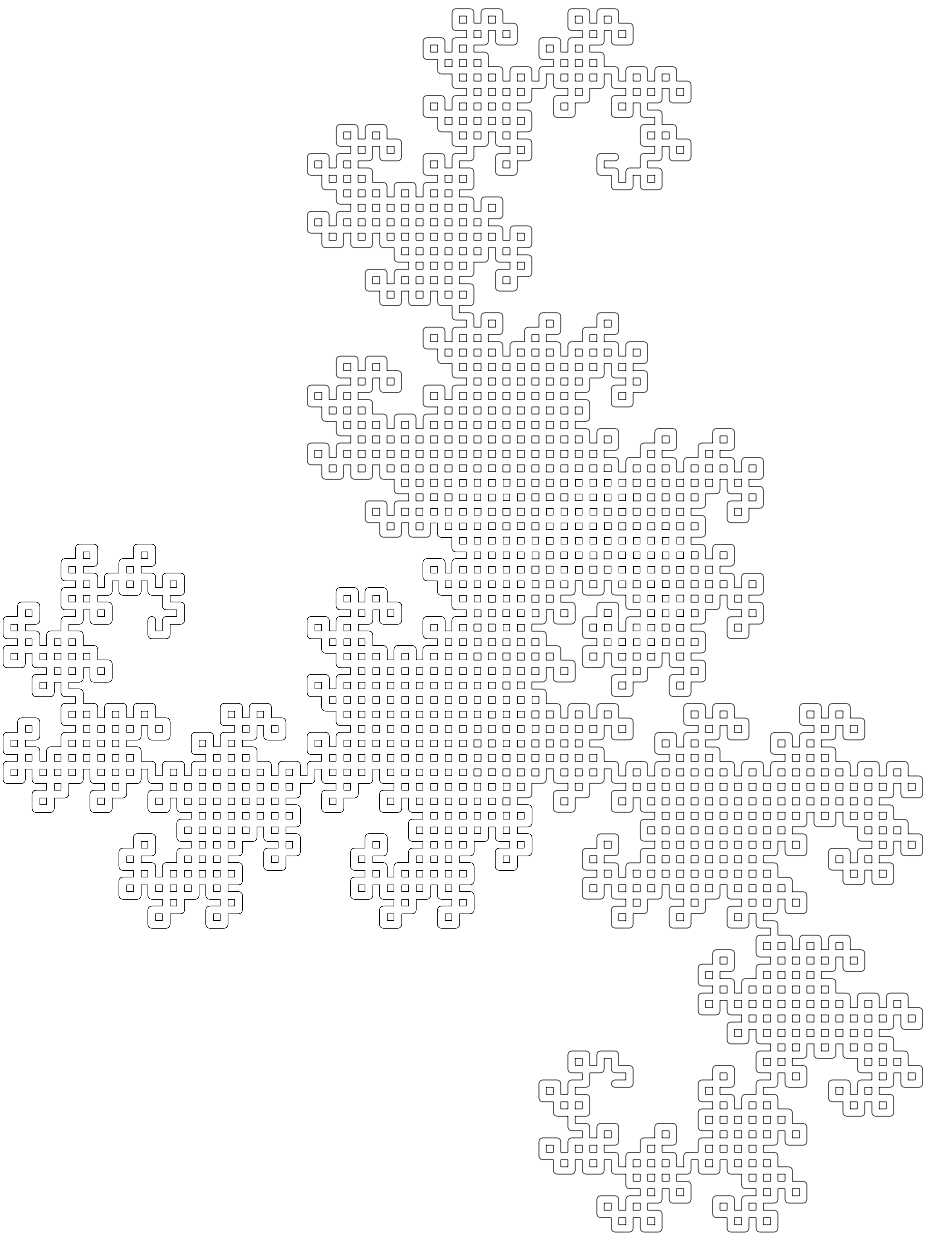
Peano-Gosper Curve

L-System Generator • 4 Iterations



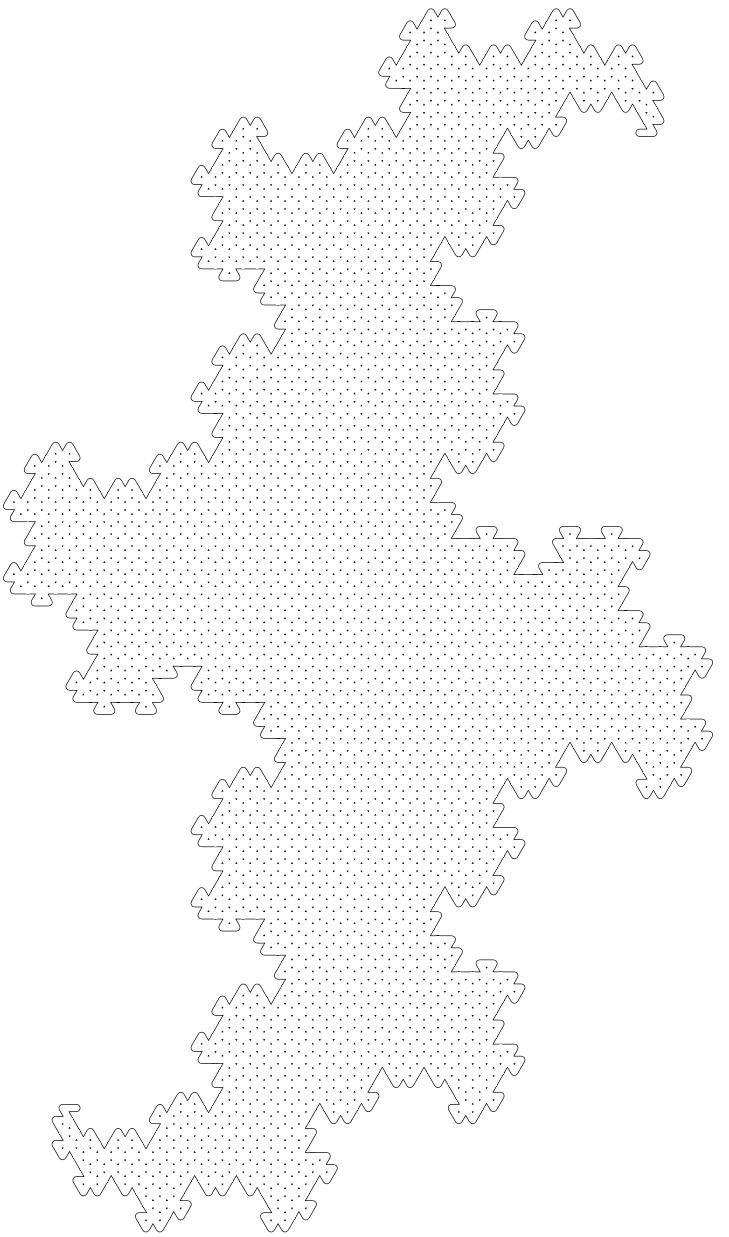
Dragon Curve

L-System Generator • 12 Iterations



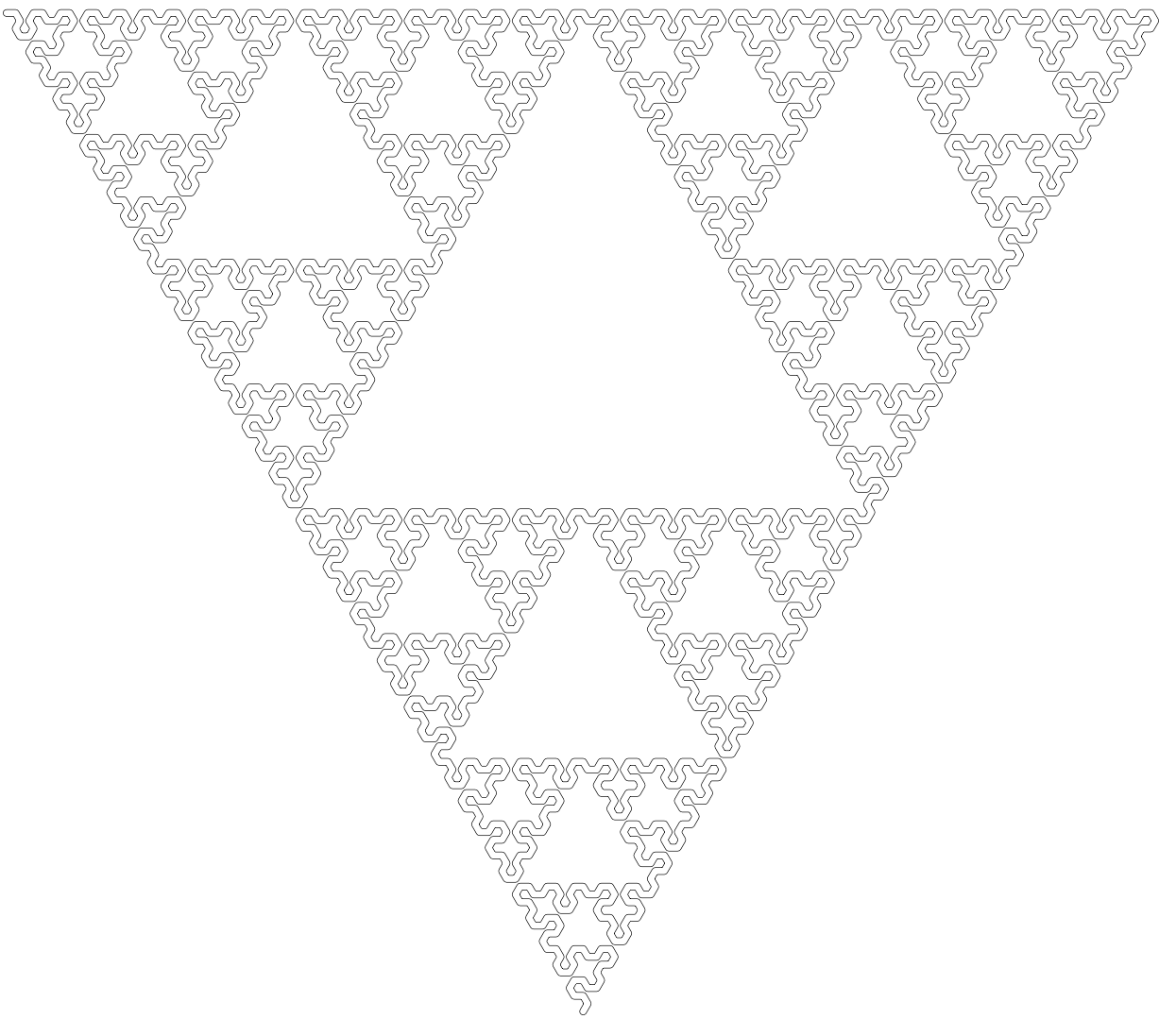
Twin Dragon

L-System Generator • 11 Iterations



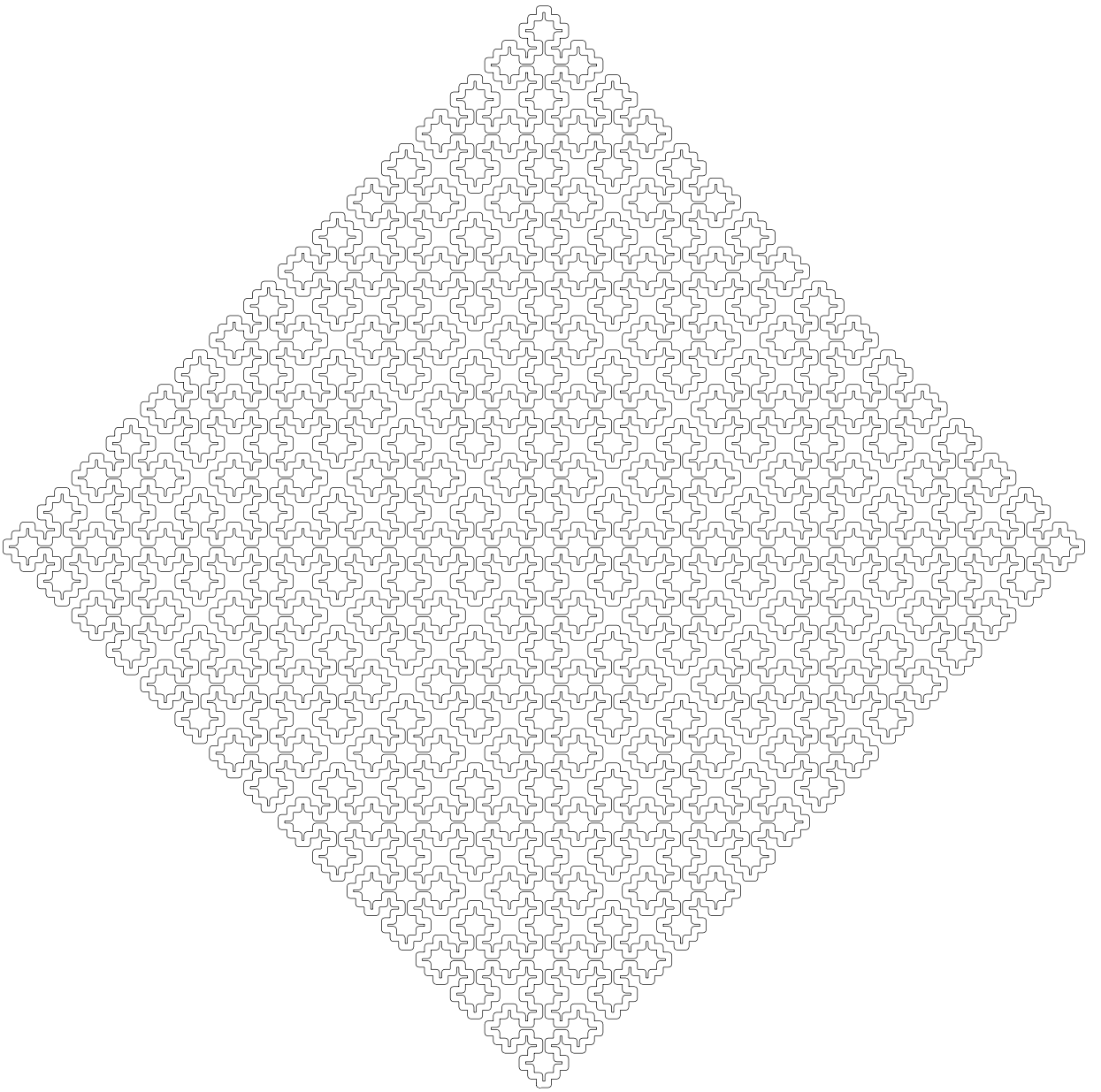
Terdragon

L-System Generator • 8 Iterations



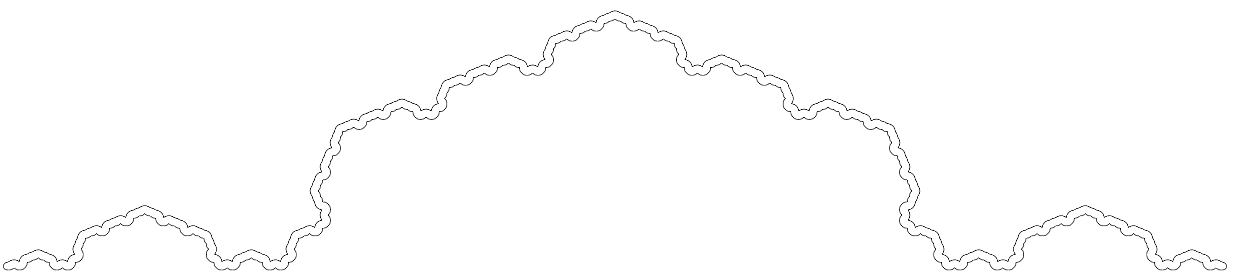
Sierpinski Arrowhead

L-System Generator • 7 Iterations



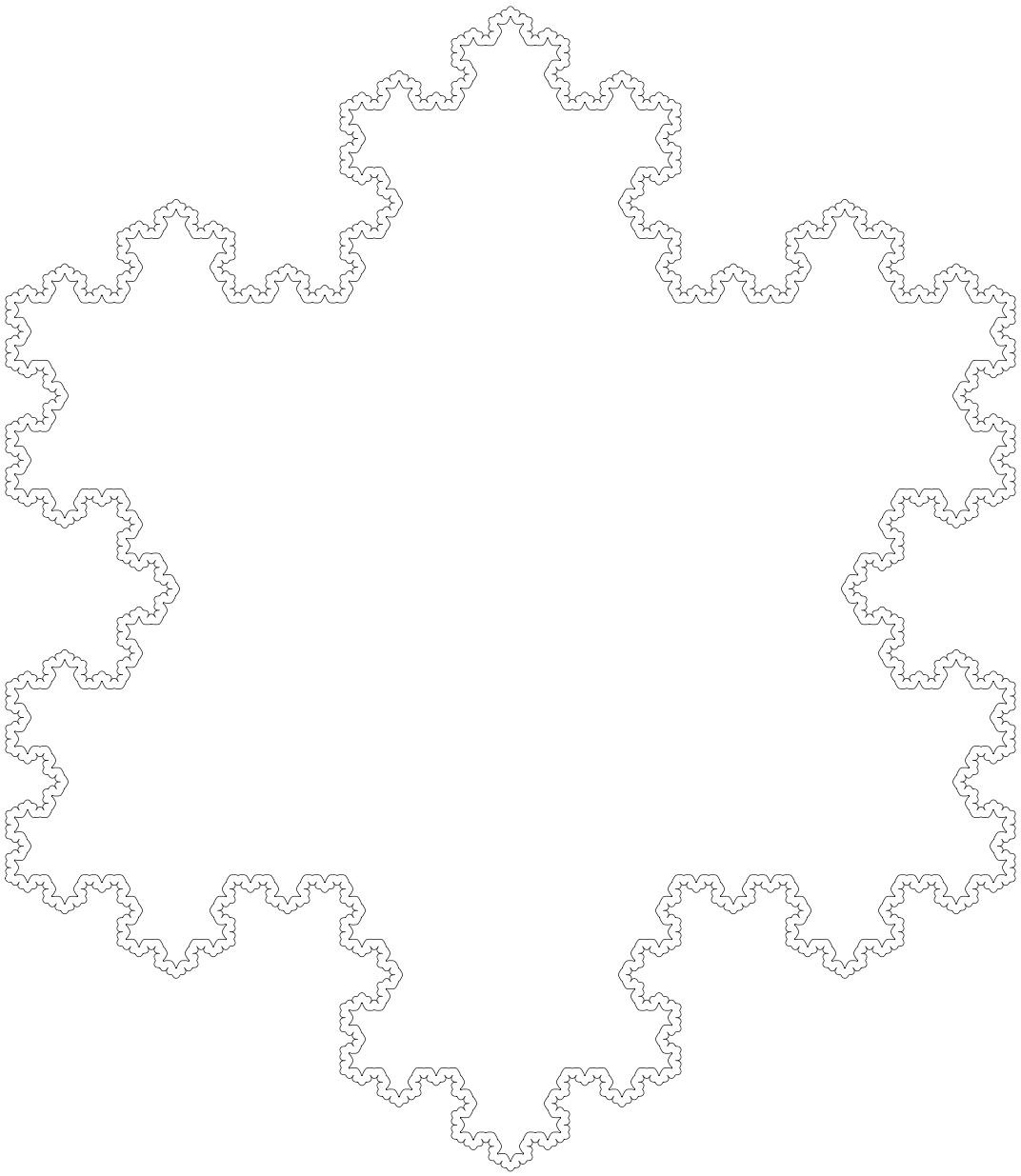
Sierpinski Curve

L-System Generator • 5 Iterations



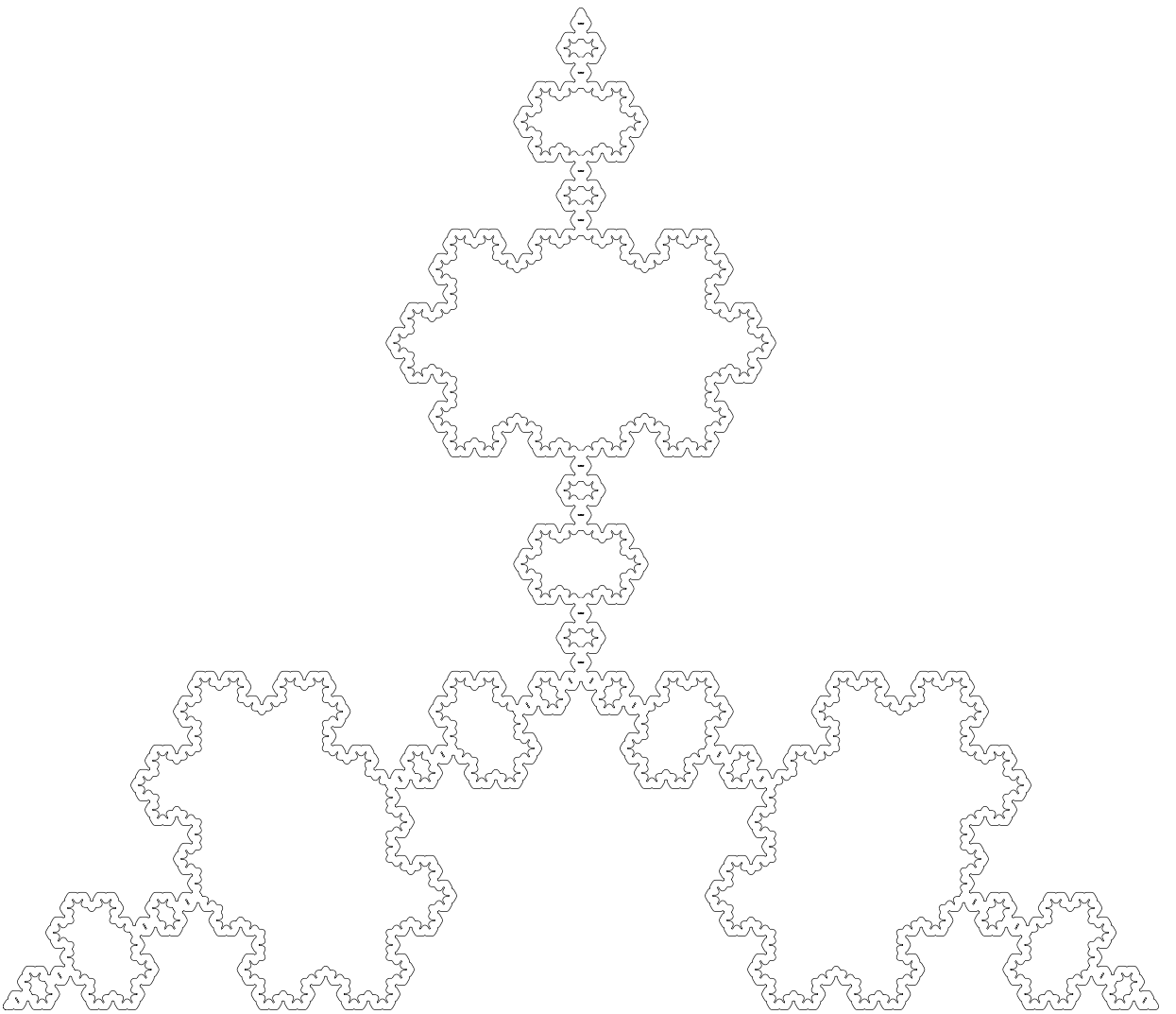
Lebesgue Curve

L-System Generator • 5 Iterations



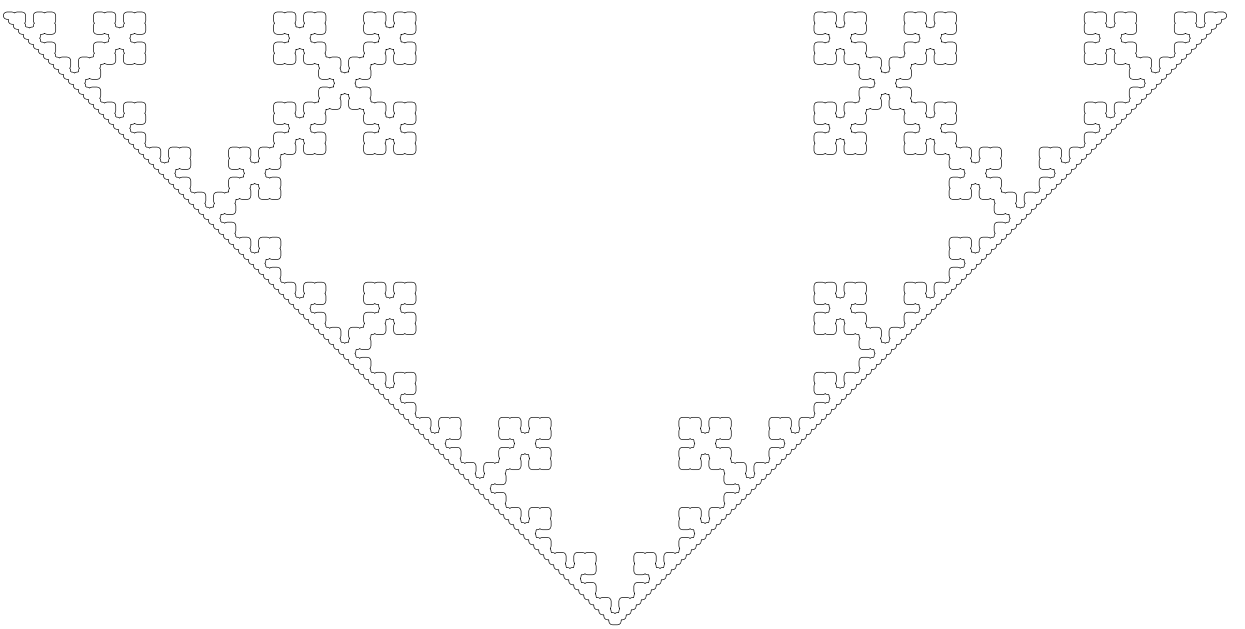
Koch Snowflake

L-System Generator • 5 Iterations



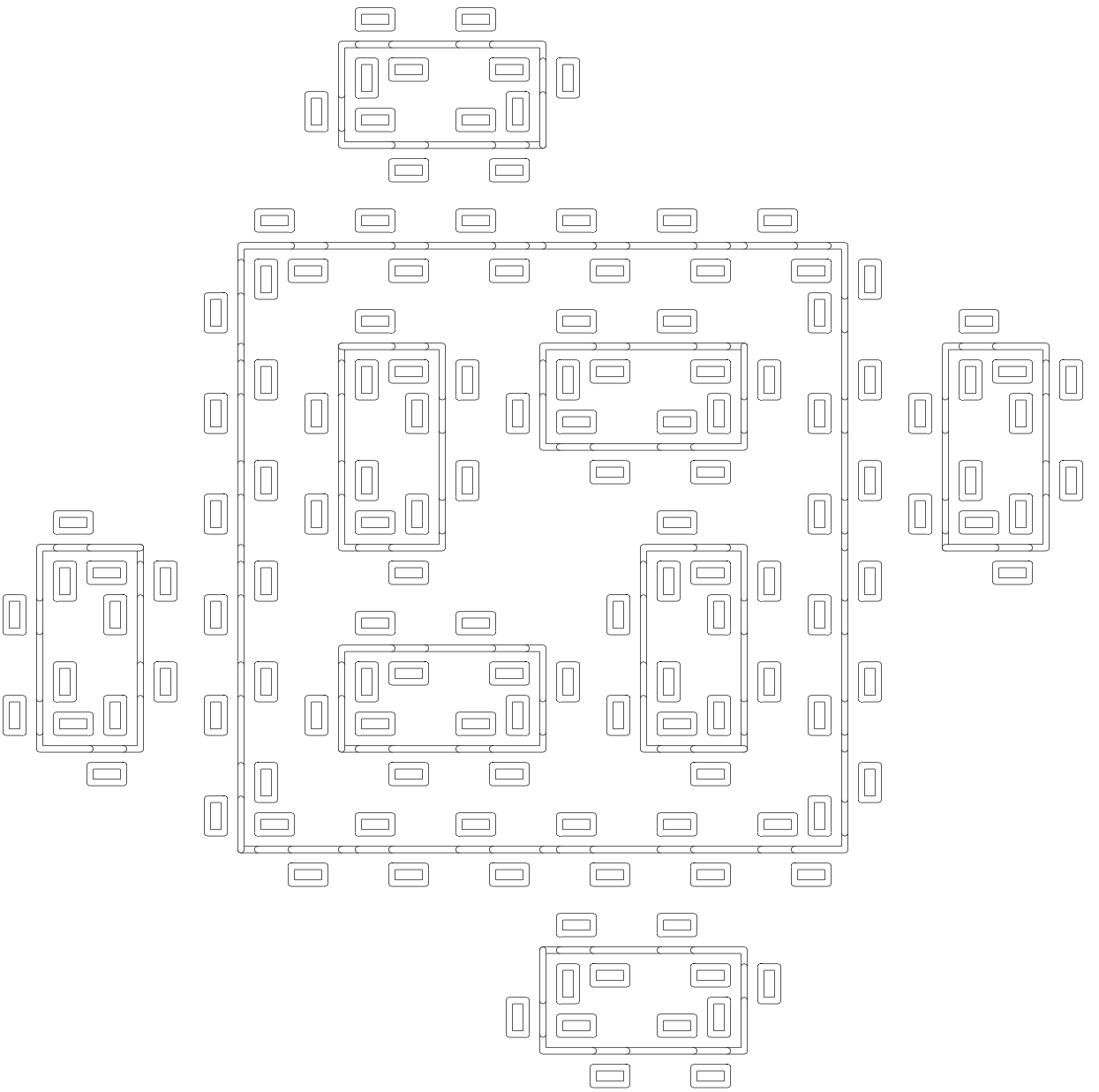
Koch Anti-Snowflake

L-System Generator • 5 Iterations



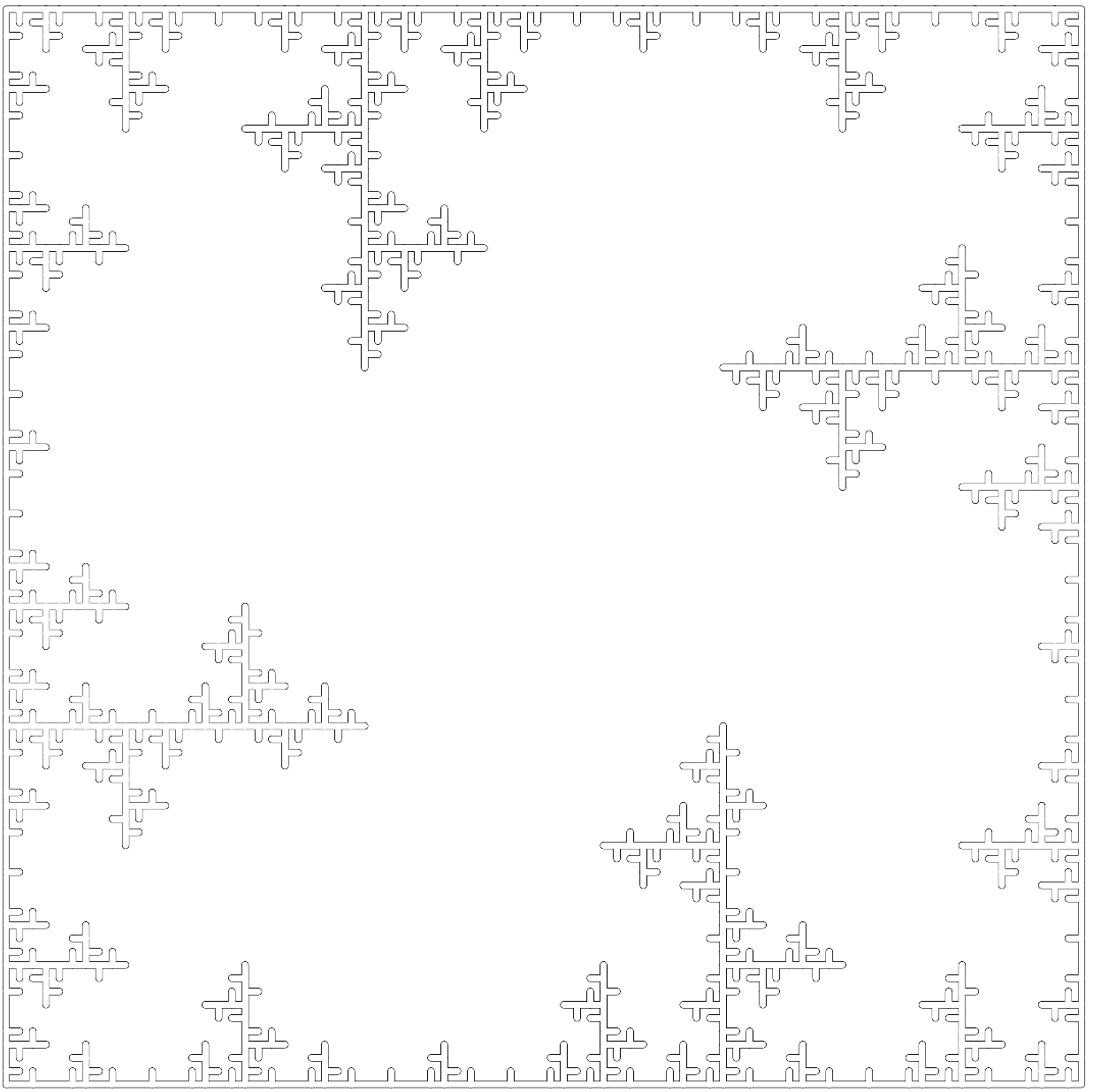
Quadratic Snowflake

L-System Generator • 5 Iterations



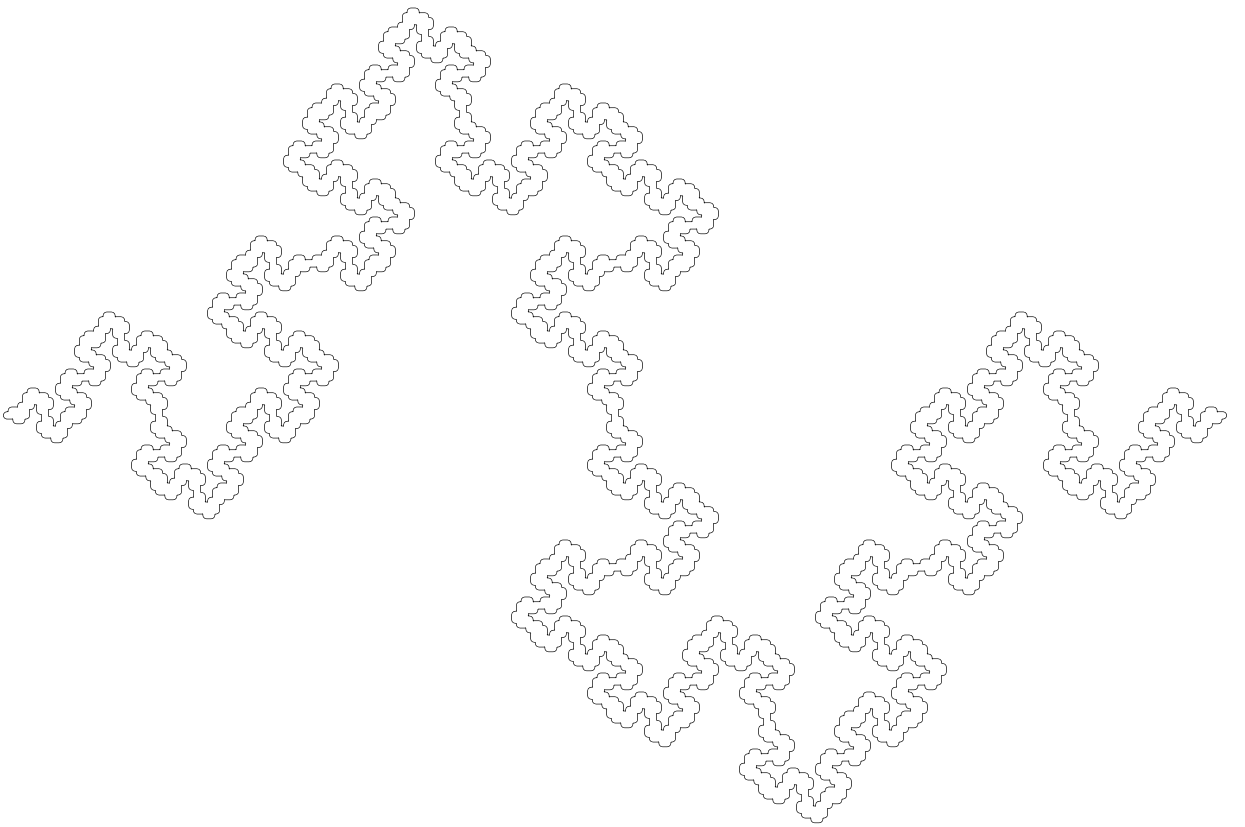
Islands and Lakes

L-System Generator • 2 Iterations



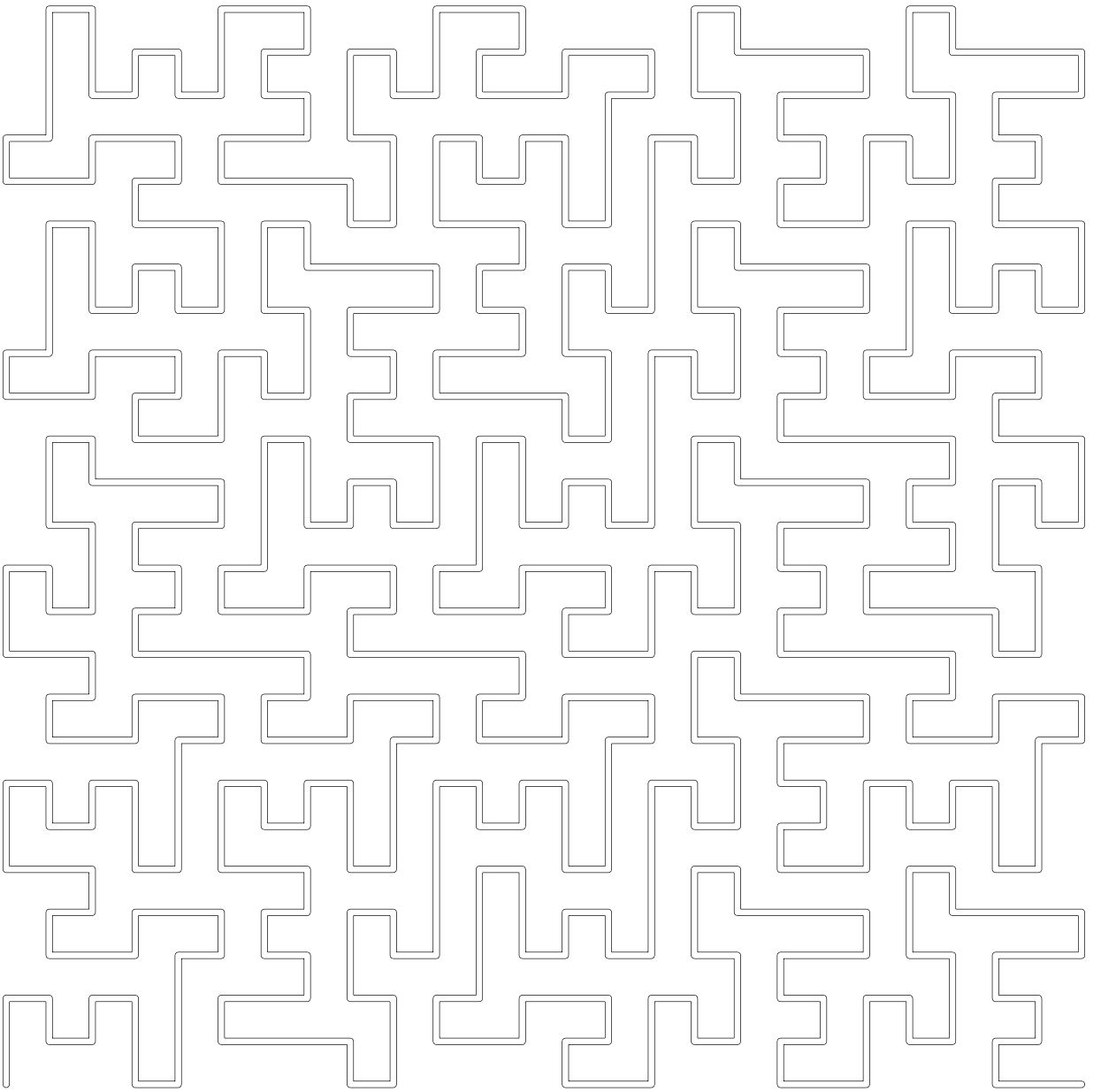
Crystal

L-System Generator • 4 Iterations



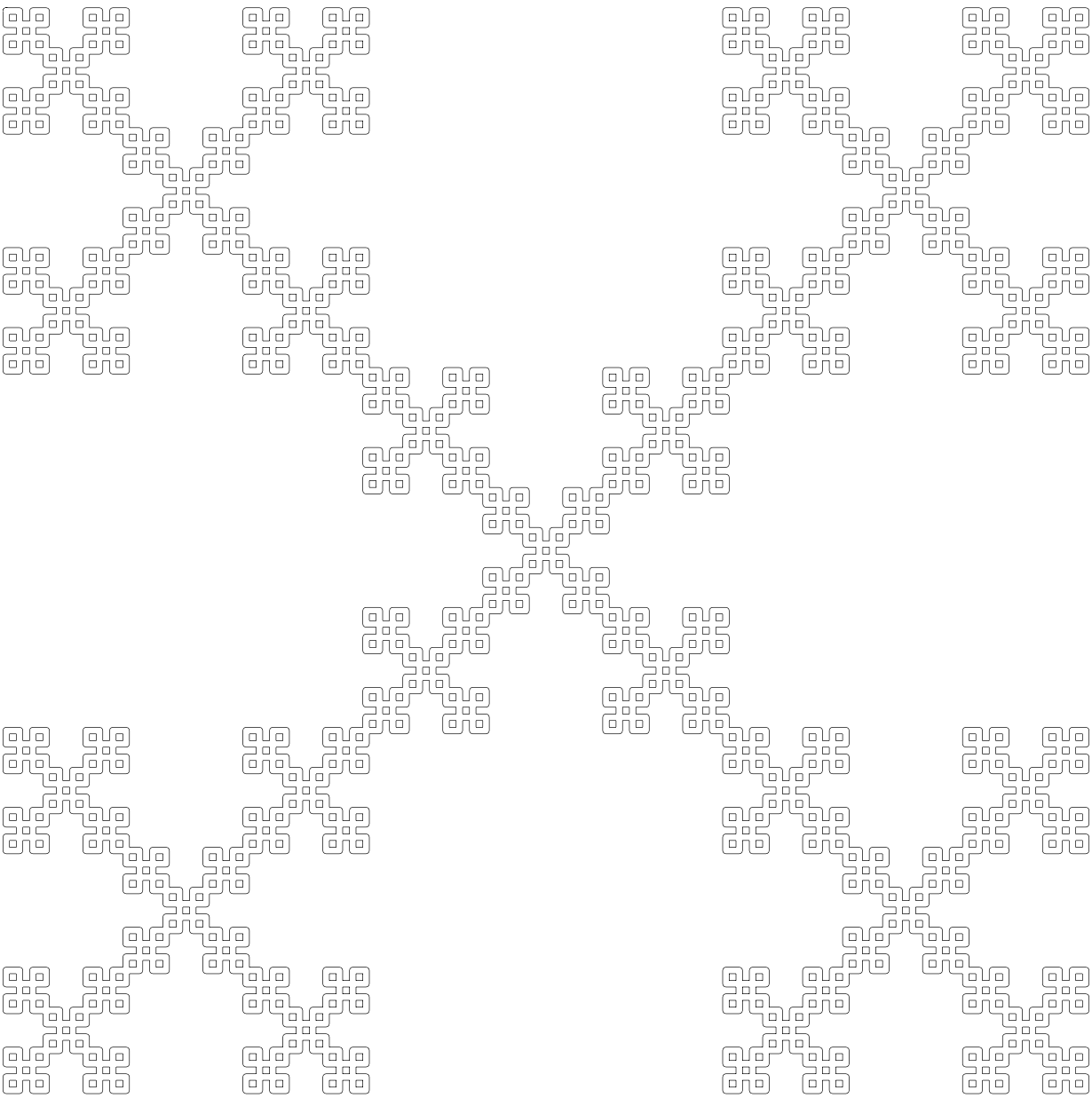
Minkowski Sausage

L-System Generator • 4 Iterations



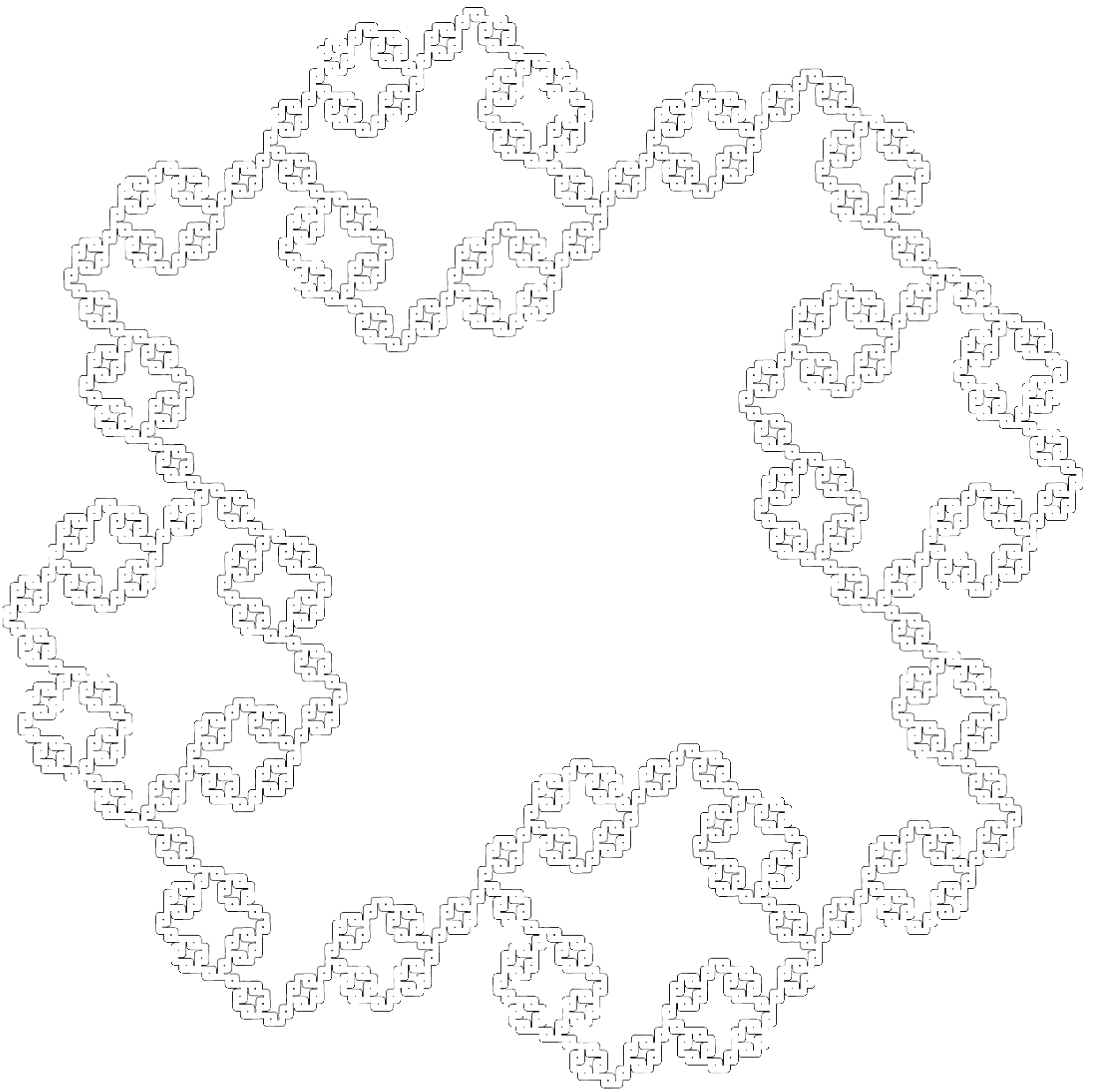
Quadratic Gosper

L-System Generator • 2 Iterations



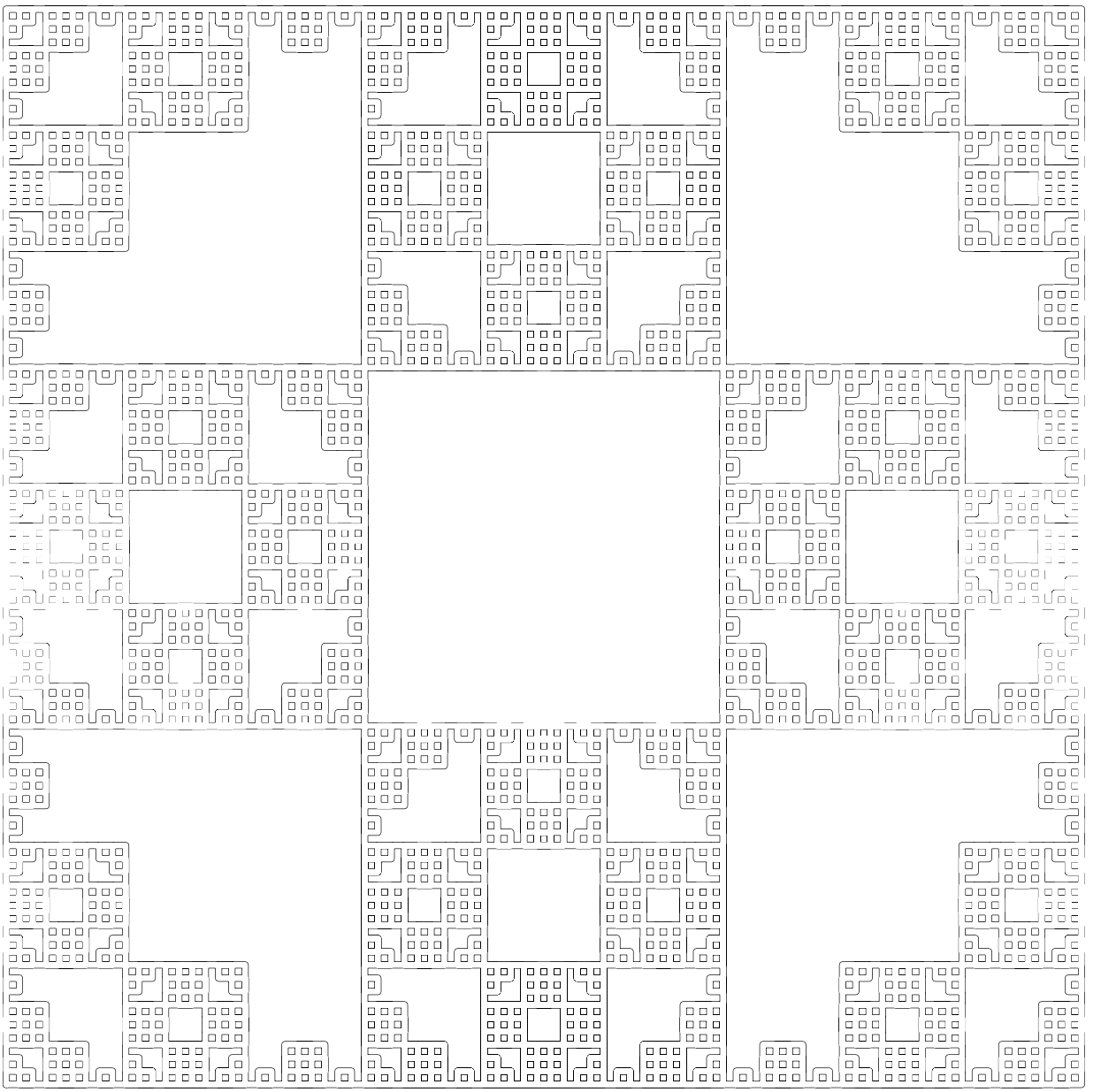
Cross Stitch

L-System Generator • 4 Iterations



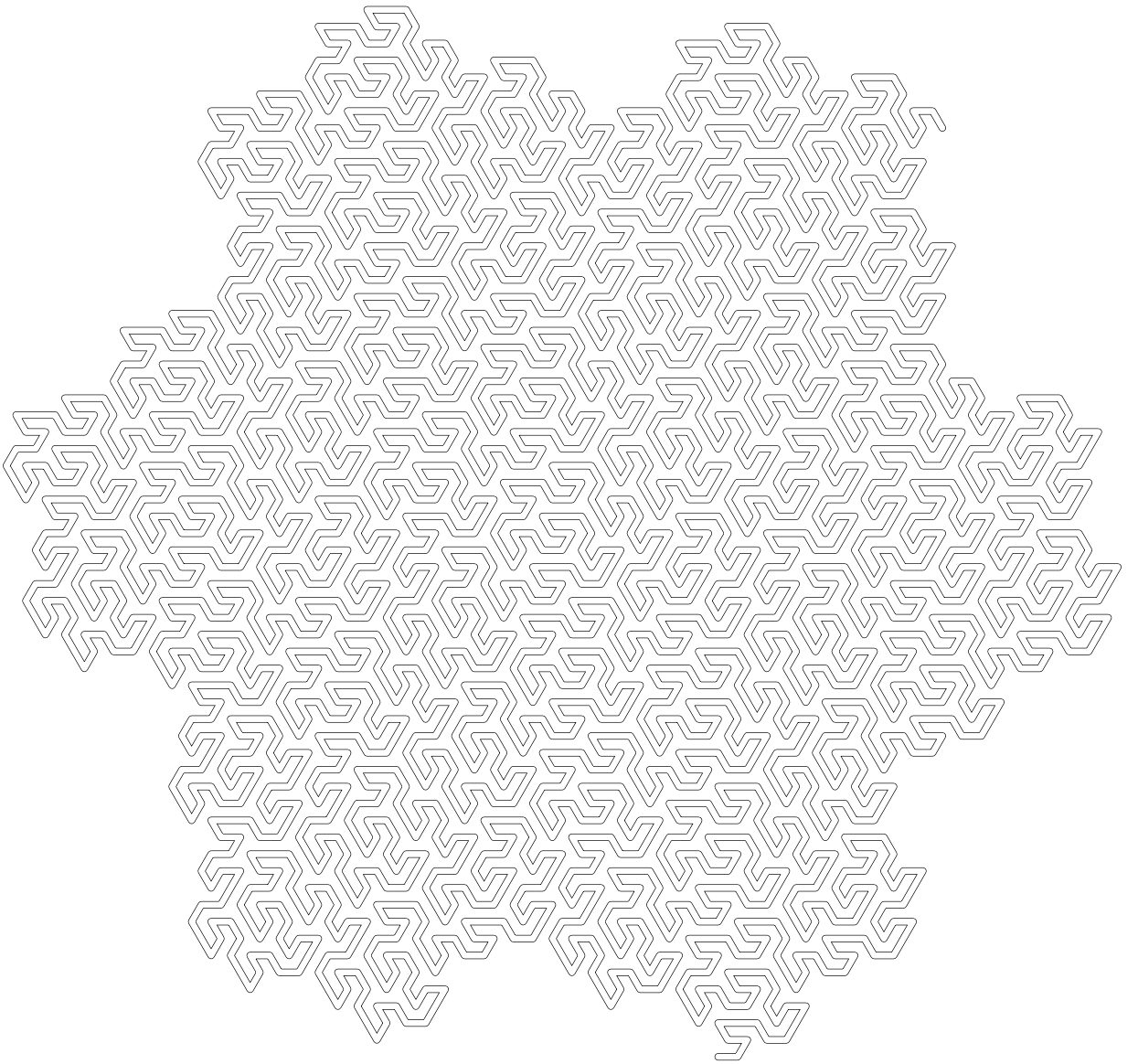
Rings

L-System Generator • 4 Iterations



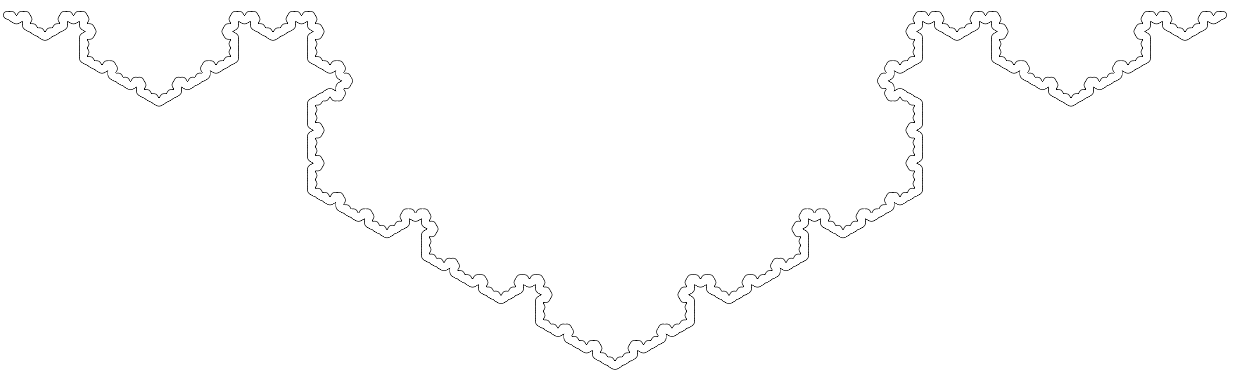
Board

L-System Generator • 4 Iterations



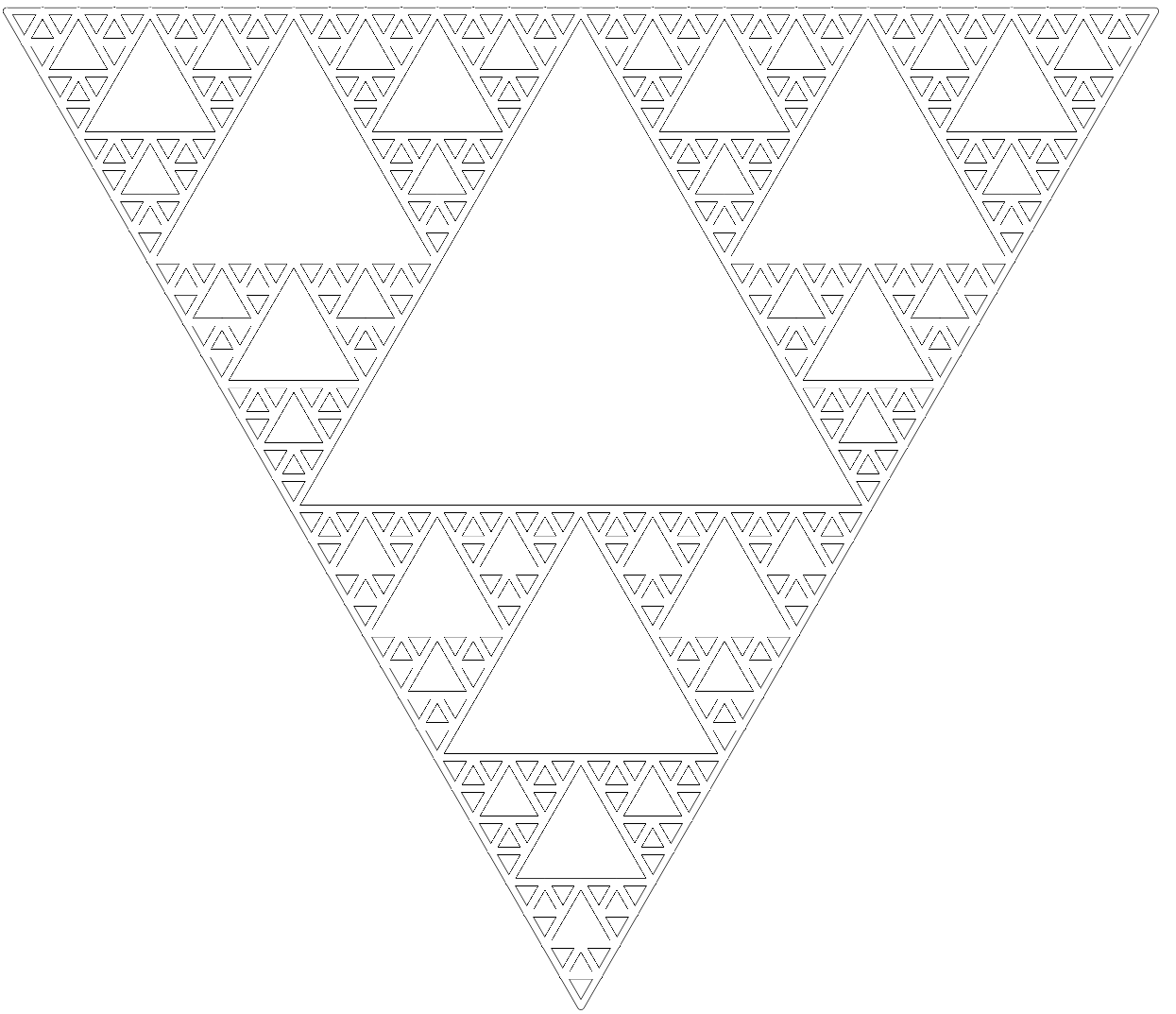
Hexagonal Gosper

L-System Generator • 4 Iterations



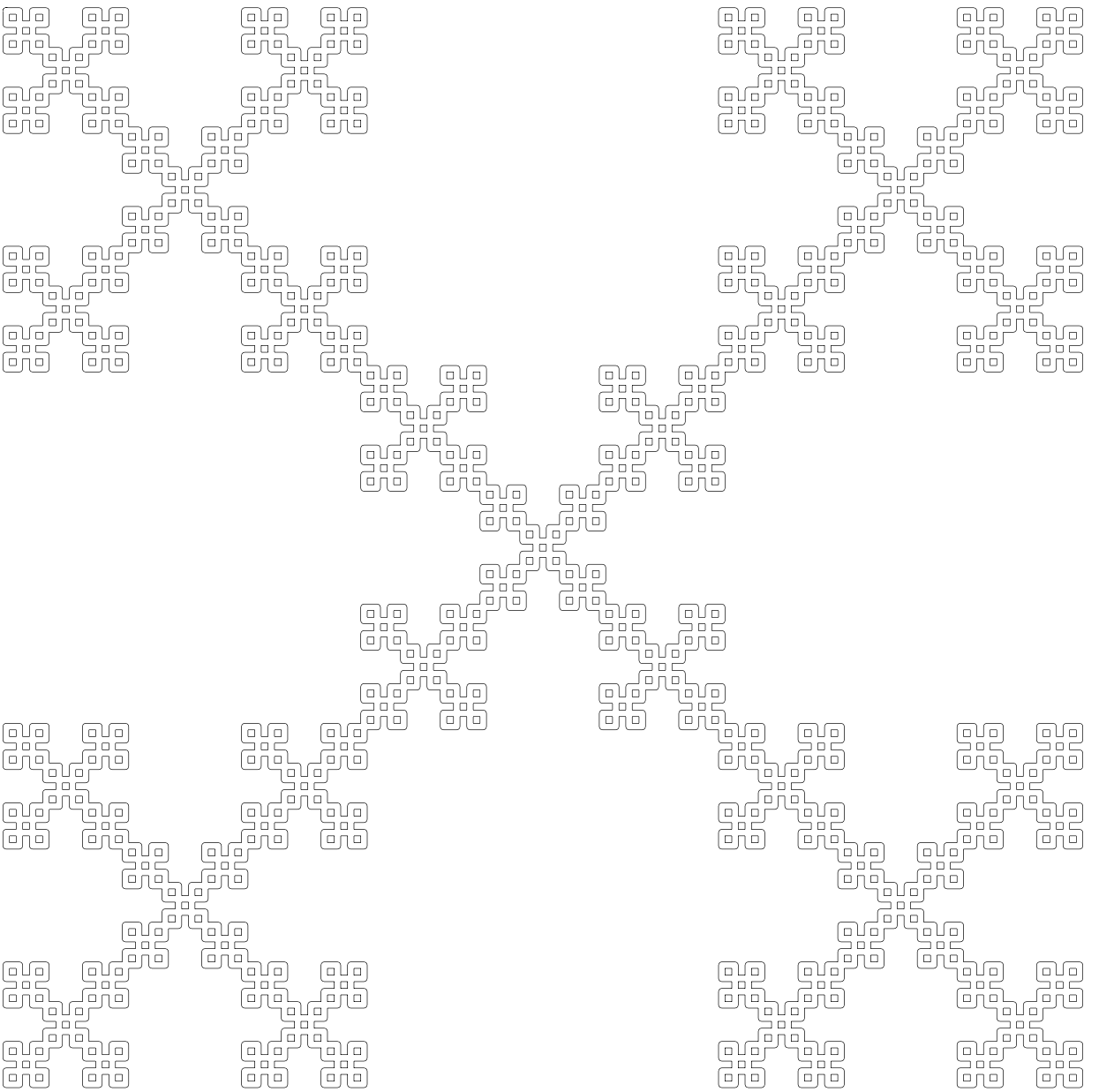
Hexagonal Tesselation

L-System Generator • 5 Iterations



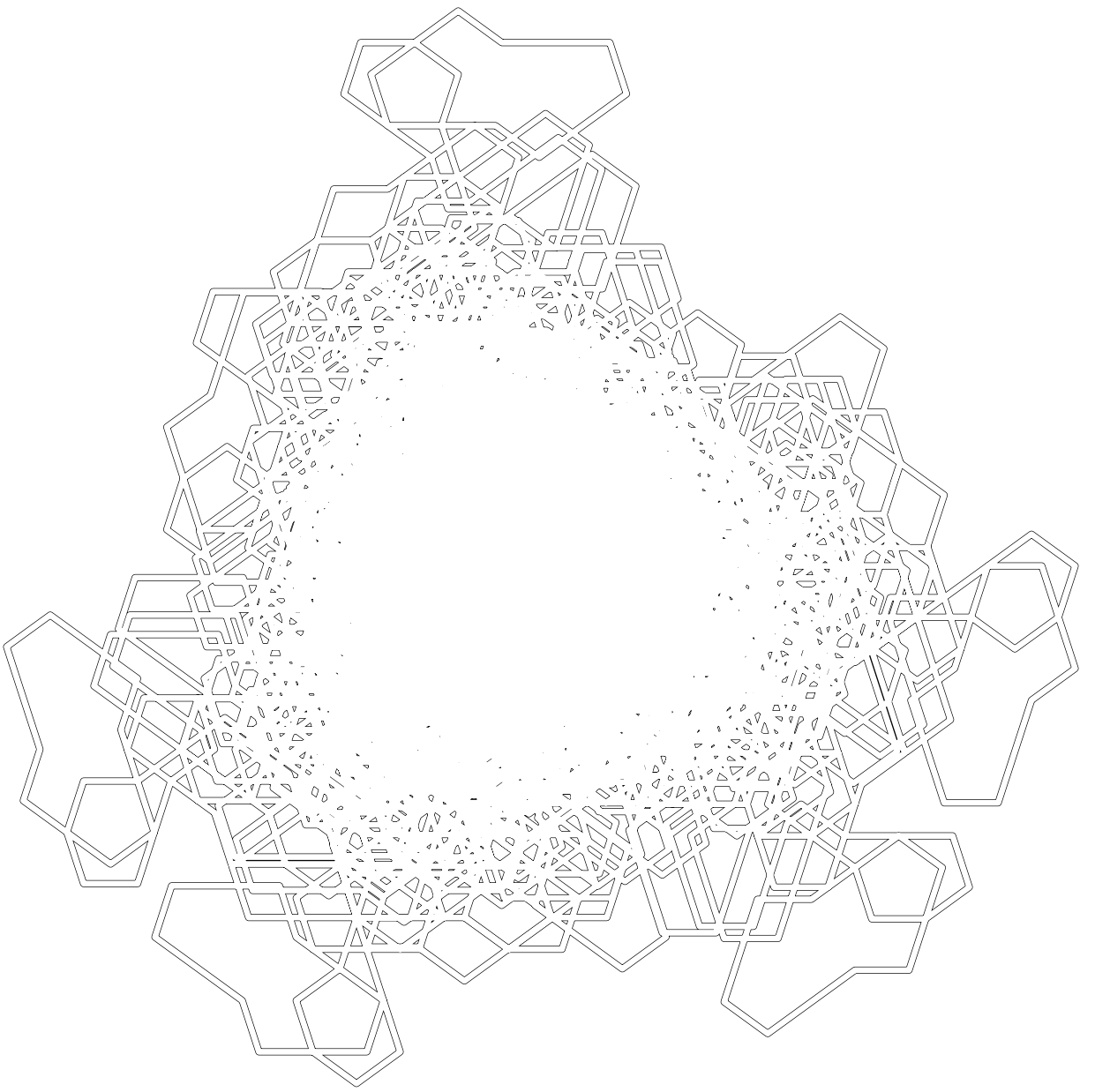
Triangle Tesselation

L-System Generator • 5 Iterations



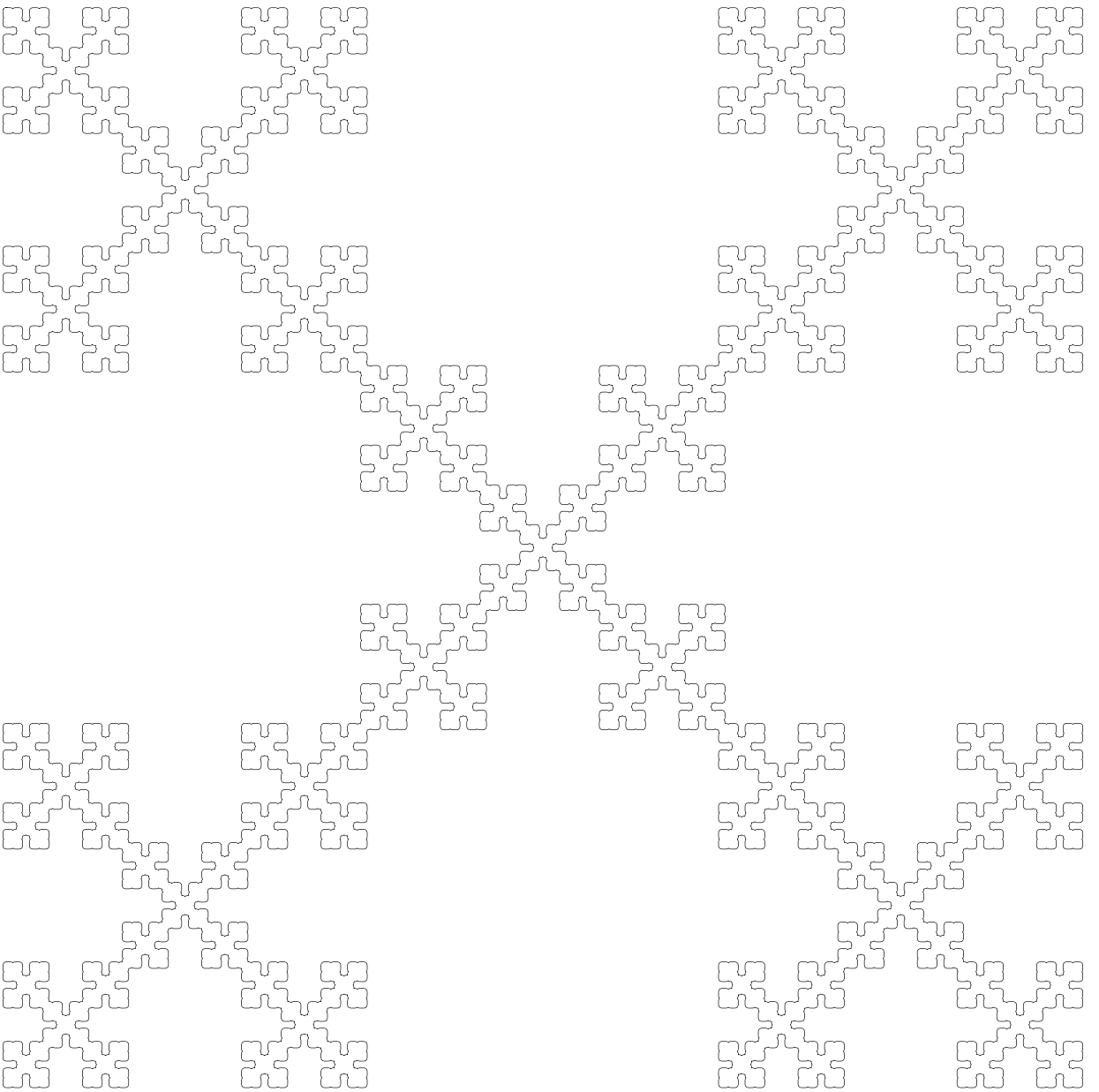
Square Tesselation

L-System Generator • 4 Iterations



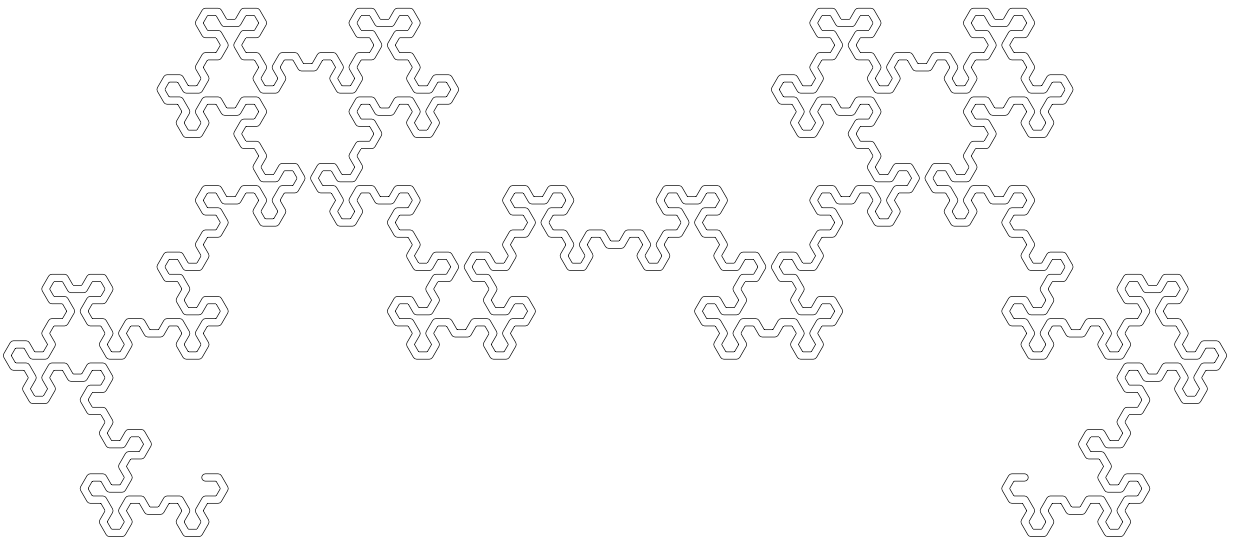
Pentaplexity

L-System Generator • 4 Iterations



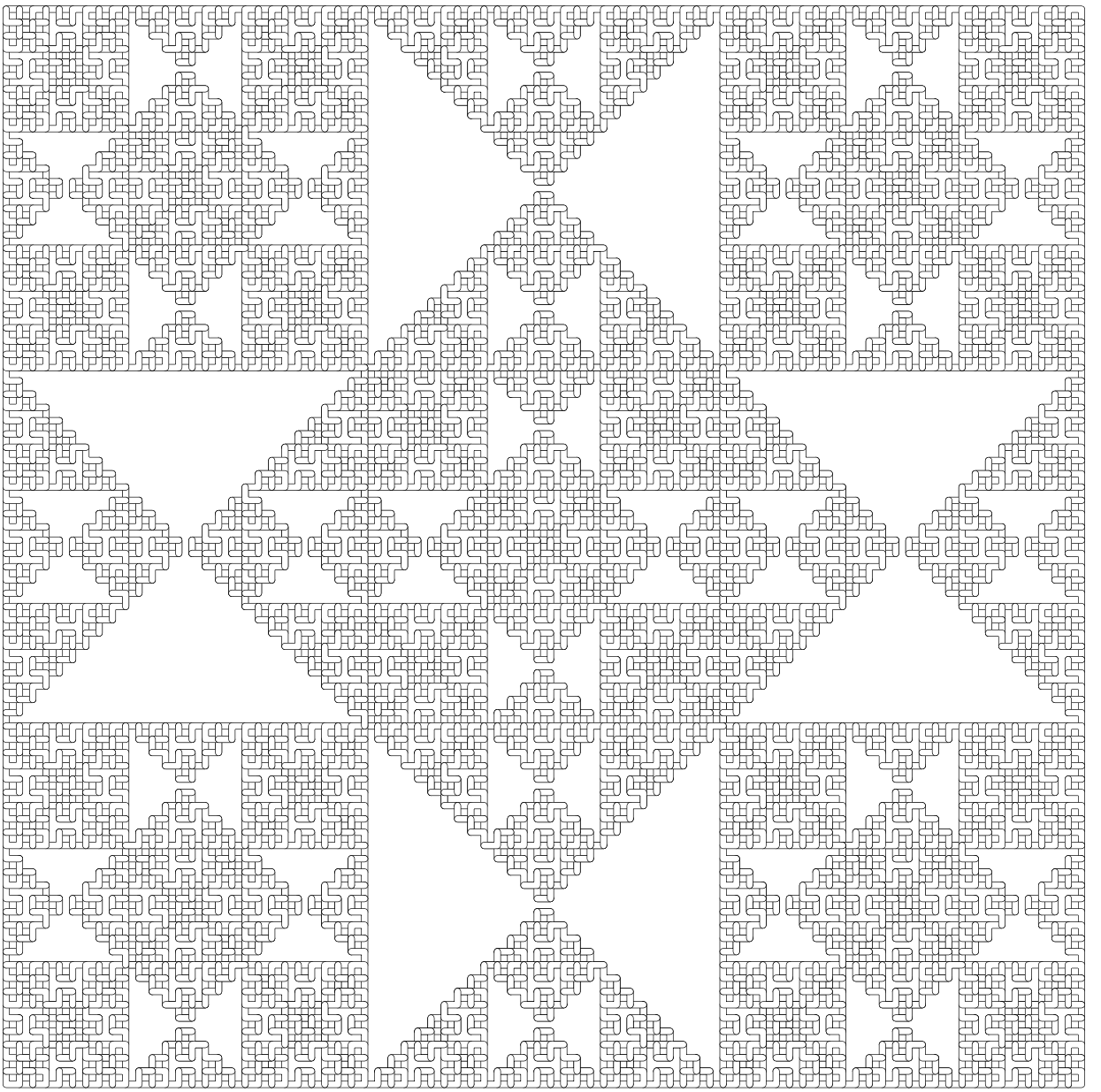
Rhombus Tesselation

L-System Generator • 5 Iterations



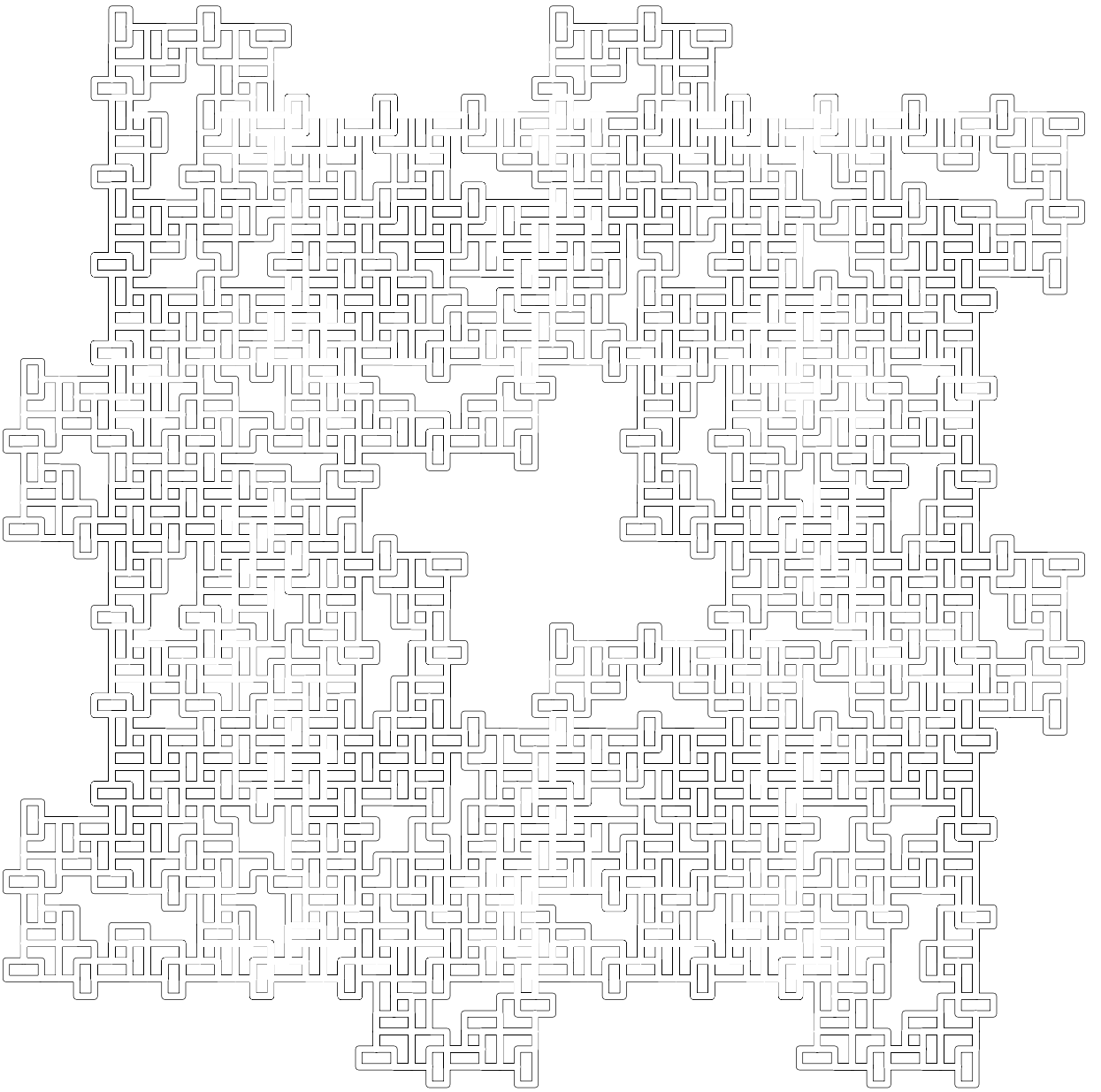
Kolam Pattern

L-System Generator • 6 Iterations



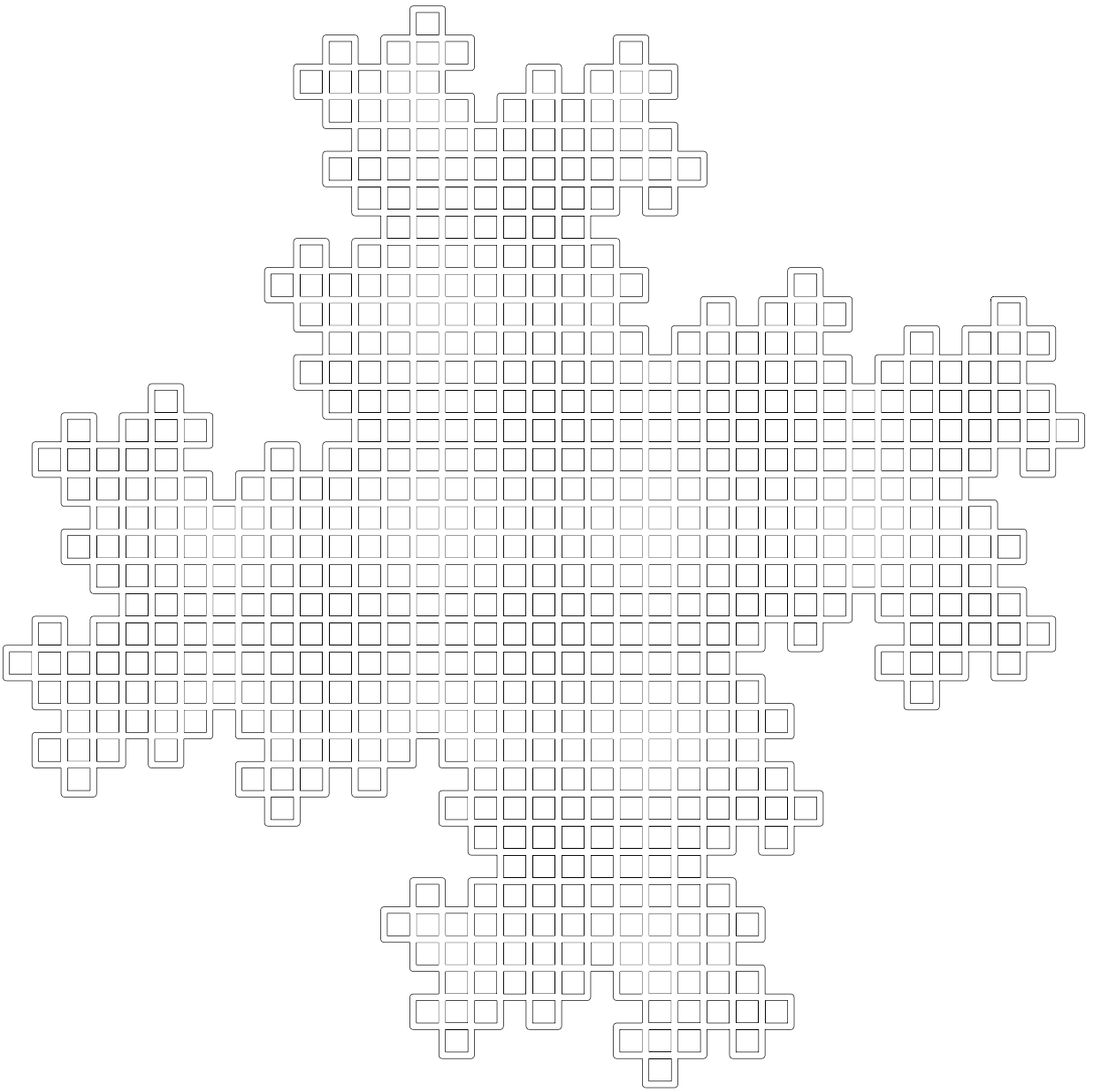
Sierpinski Carpet

L-System Generator • 4 Iterations



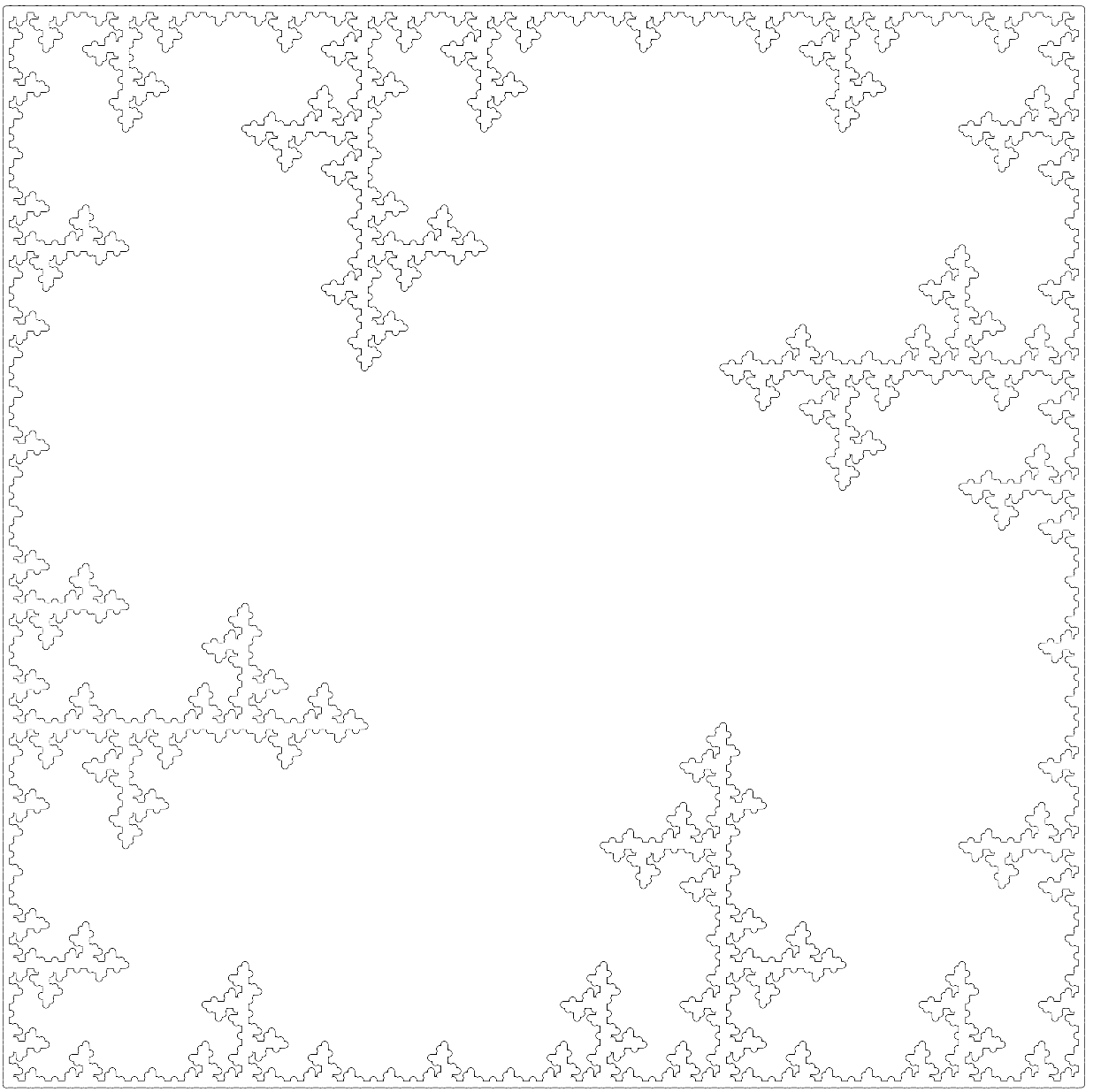
Tile Pattern

L-System Generator • 4 Iterations



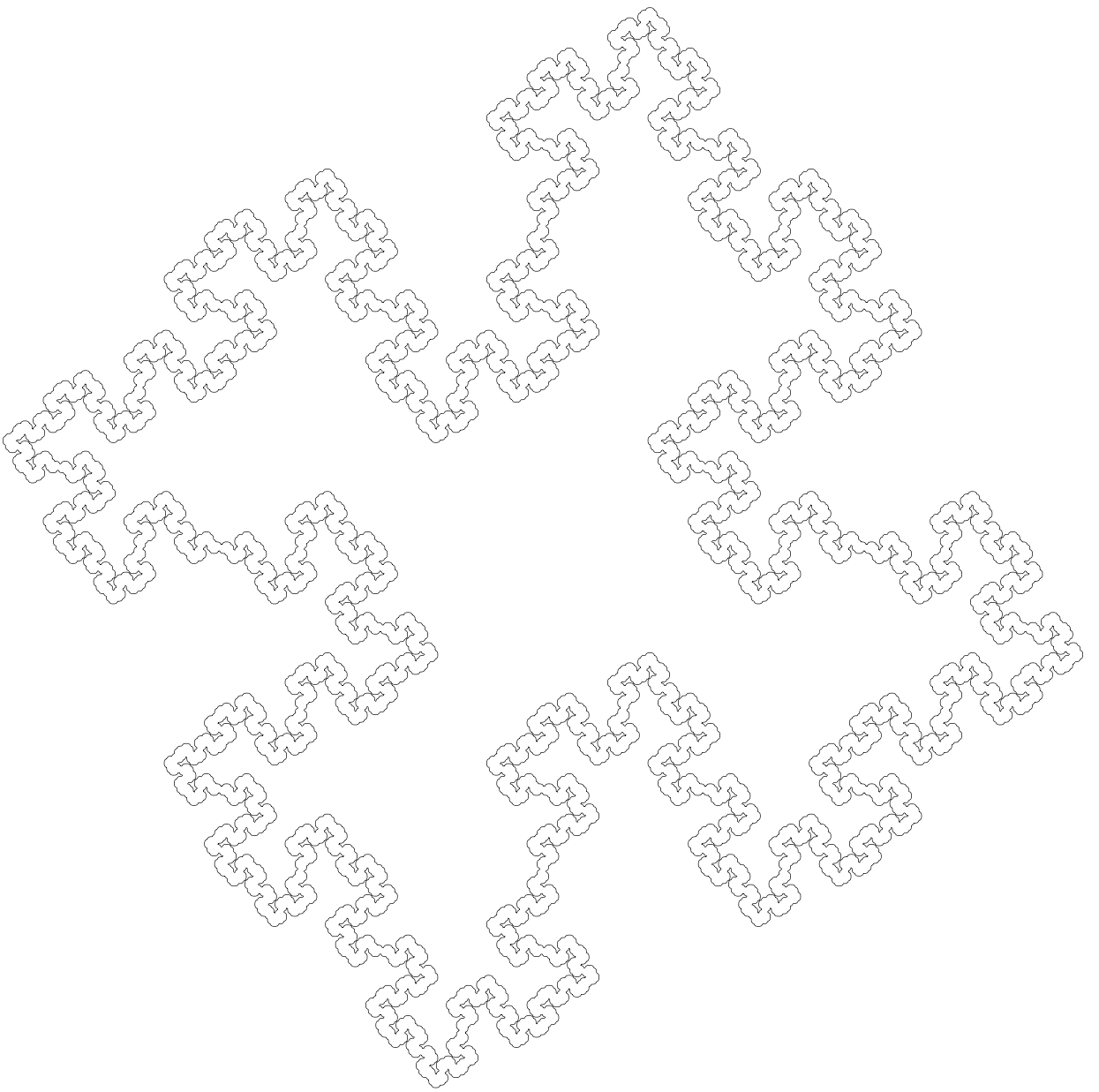
Mosaic

L-System Generator • 4 Iterations



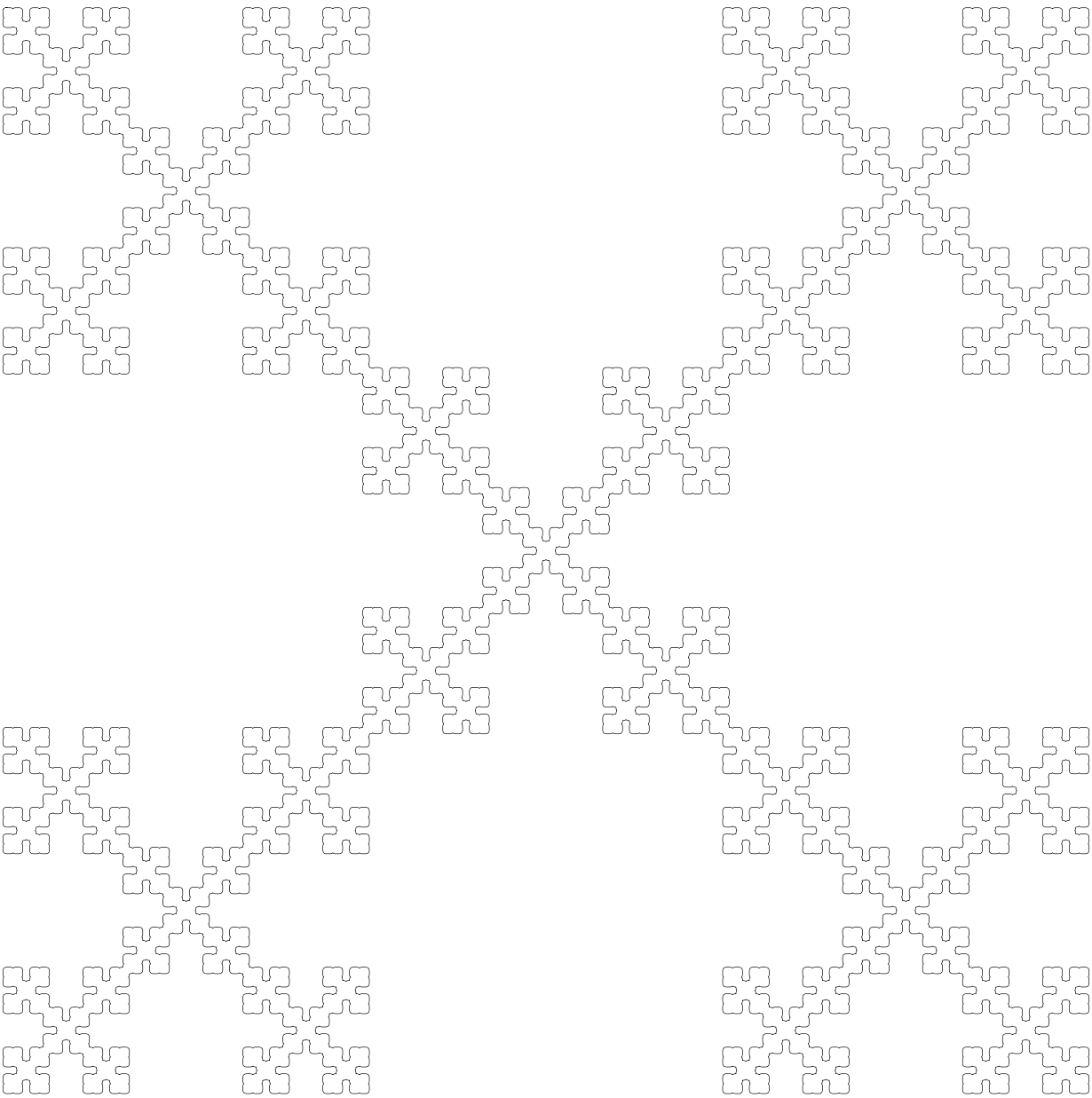
Crystal Loop

L-System Generator • 5 Iterations



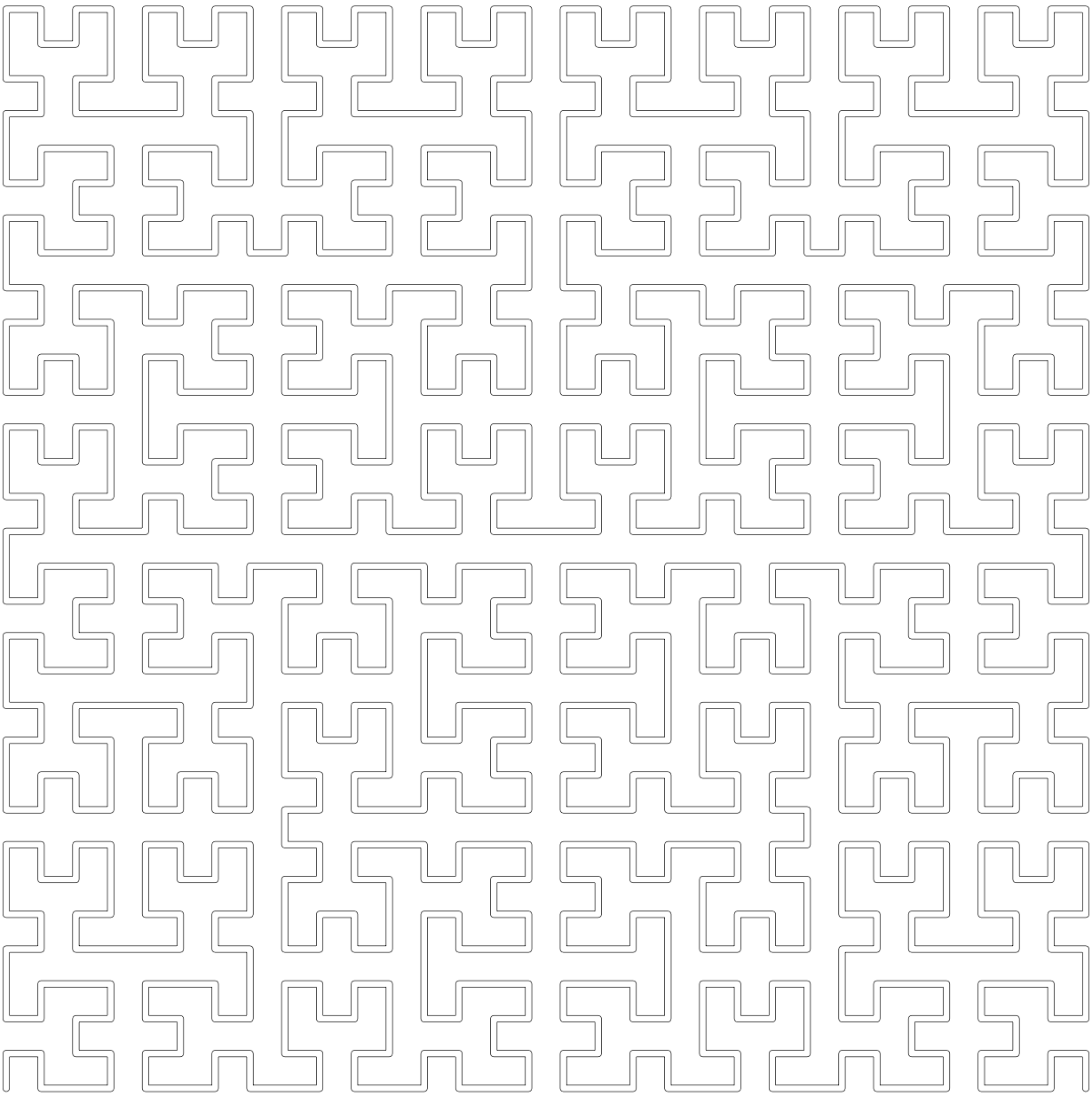
Minkowski Star

L-System Generator • 4 Iterations



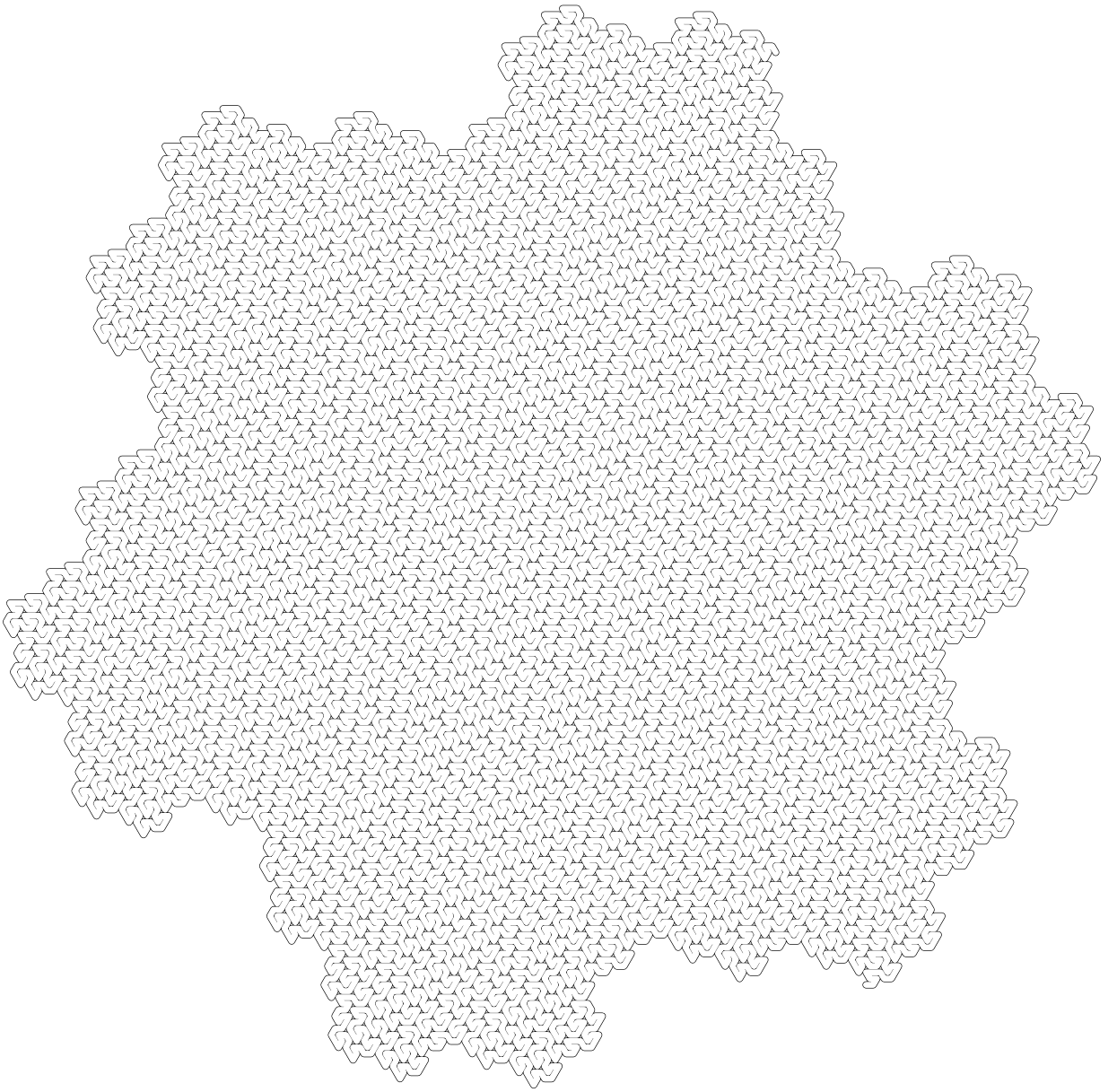
Quadratic Star

L-System Generator • 5 Iterations



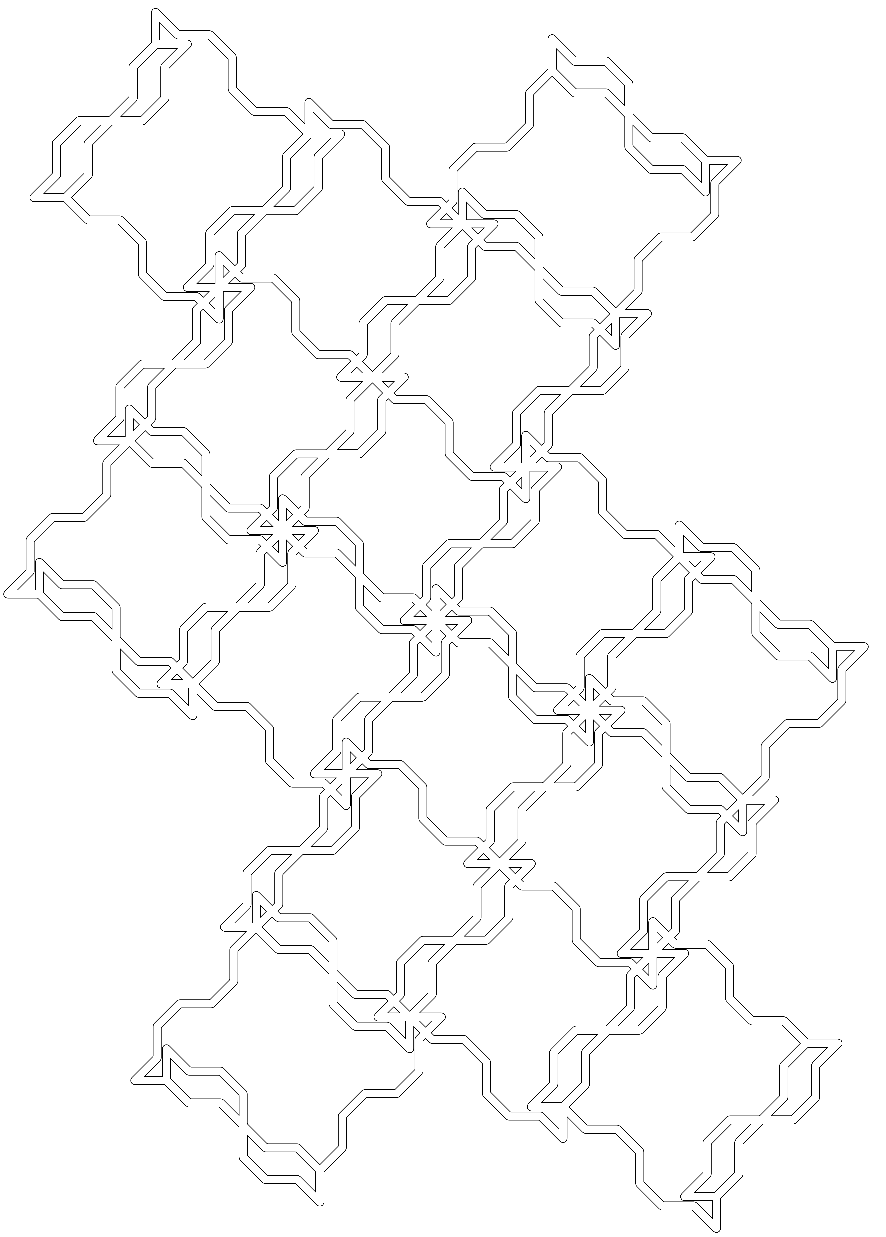
Hilbert Loop

L-System Generator • 5 Iterations



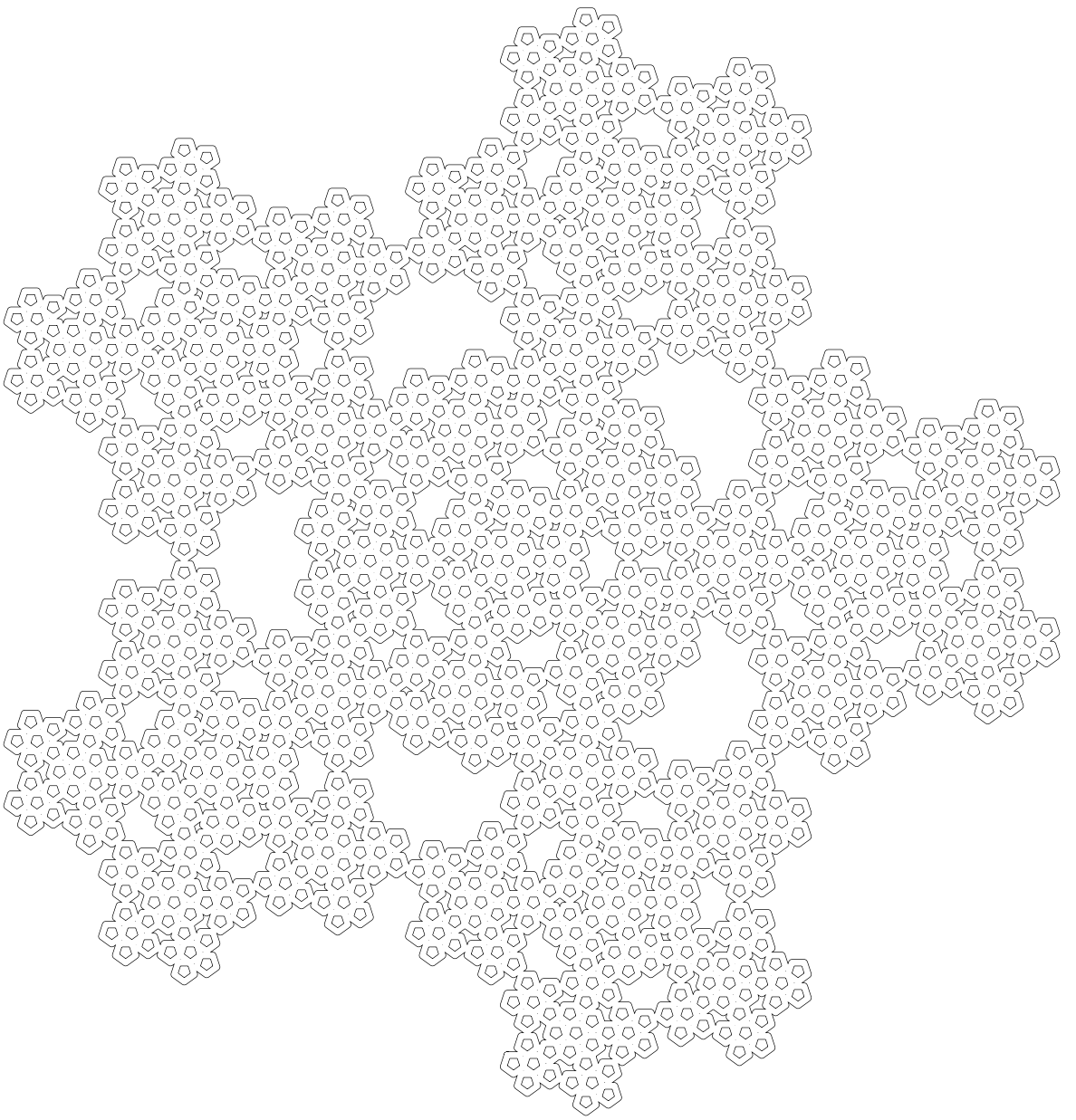
Hexagonal Maze

L-System Generator • 5 Iterations



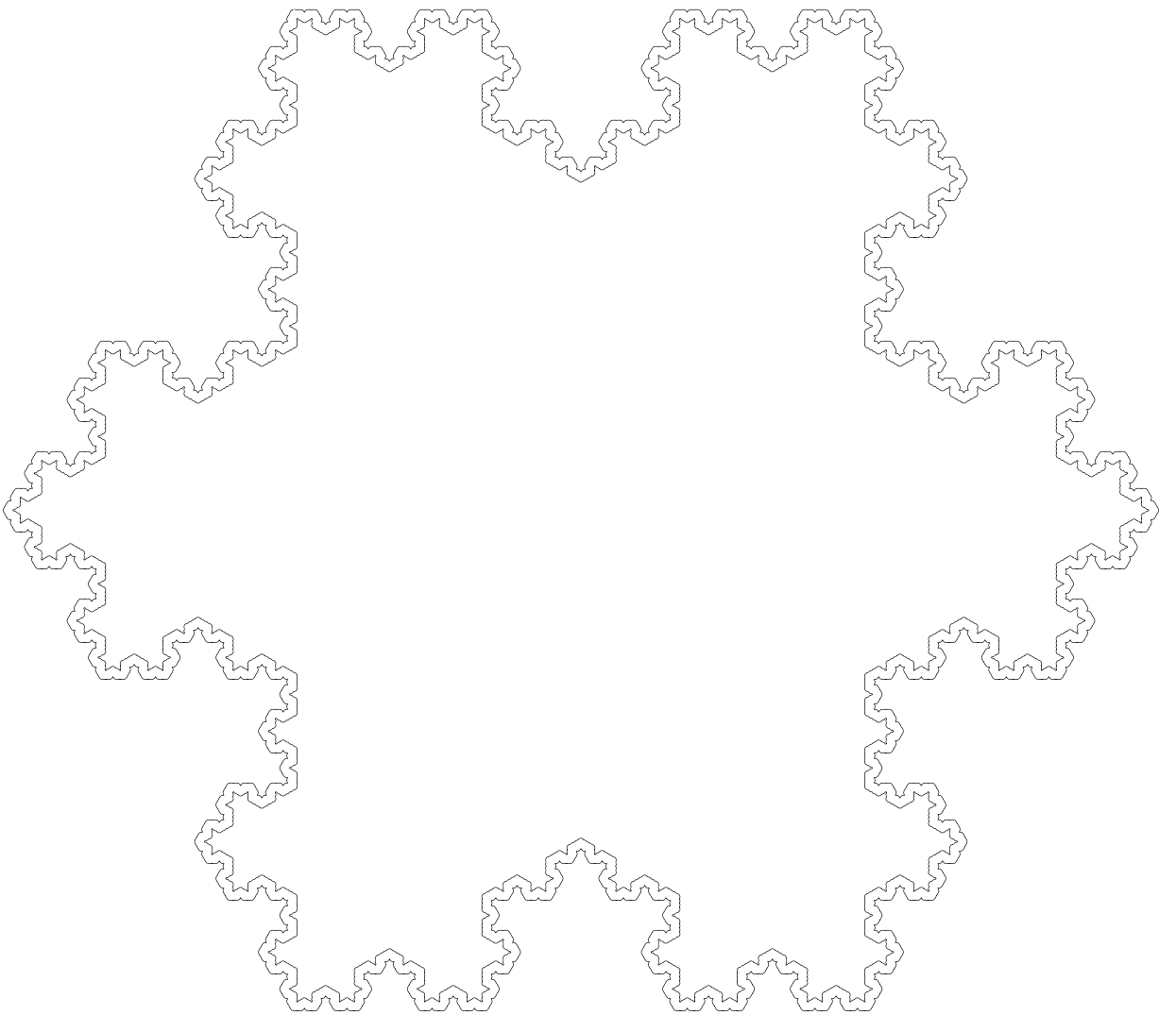
Spiderweb

L-System Generator • 3 Iterations



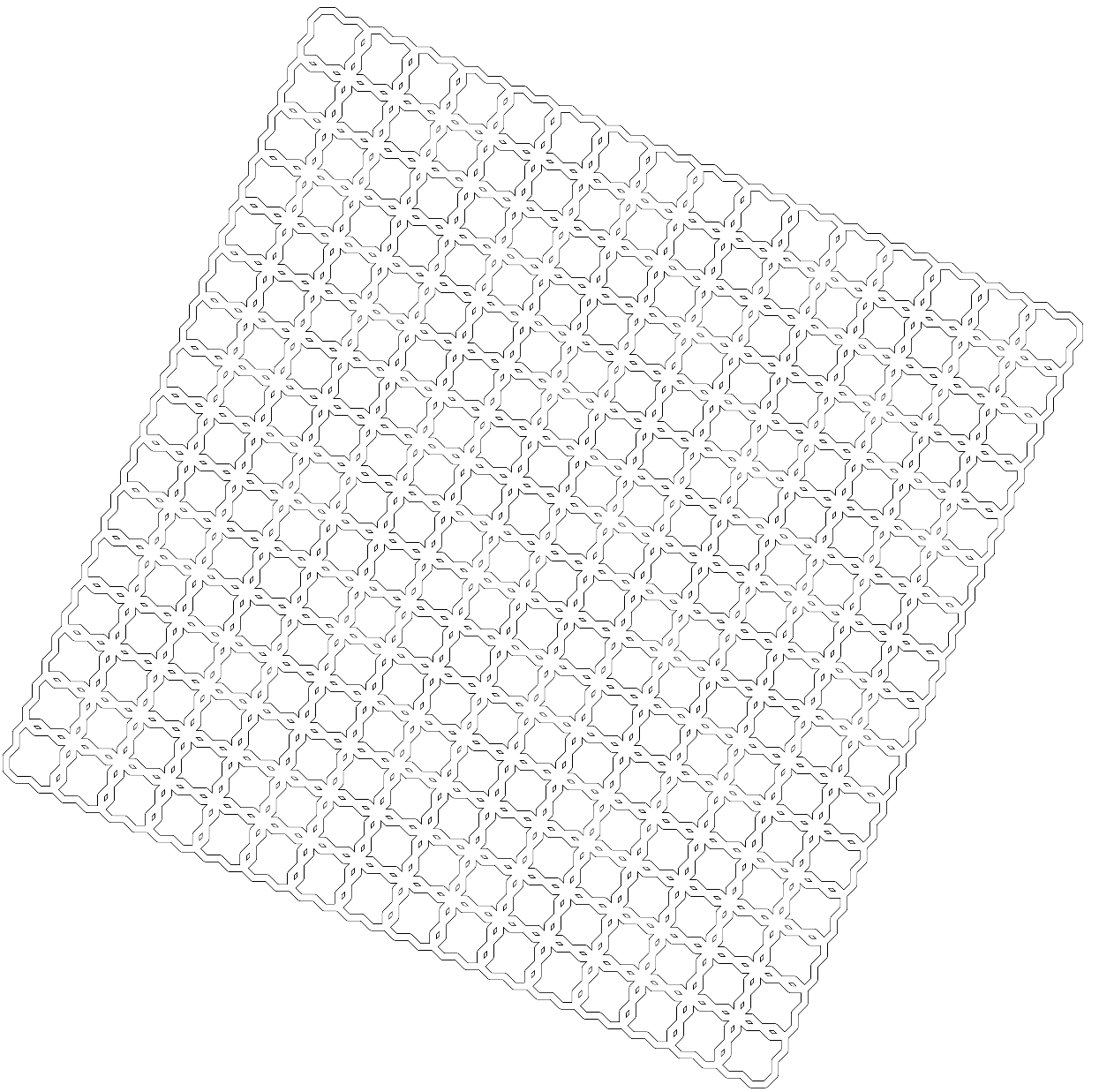
Star Tesselation

L-System Generator • 4 Iterations



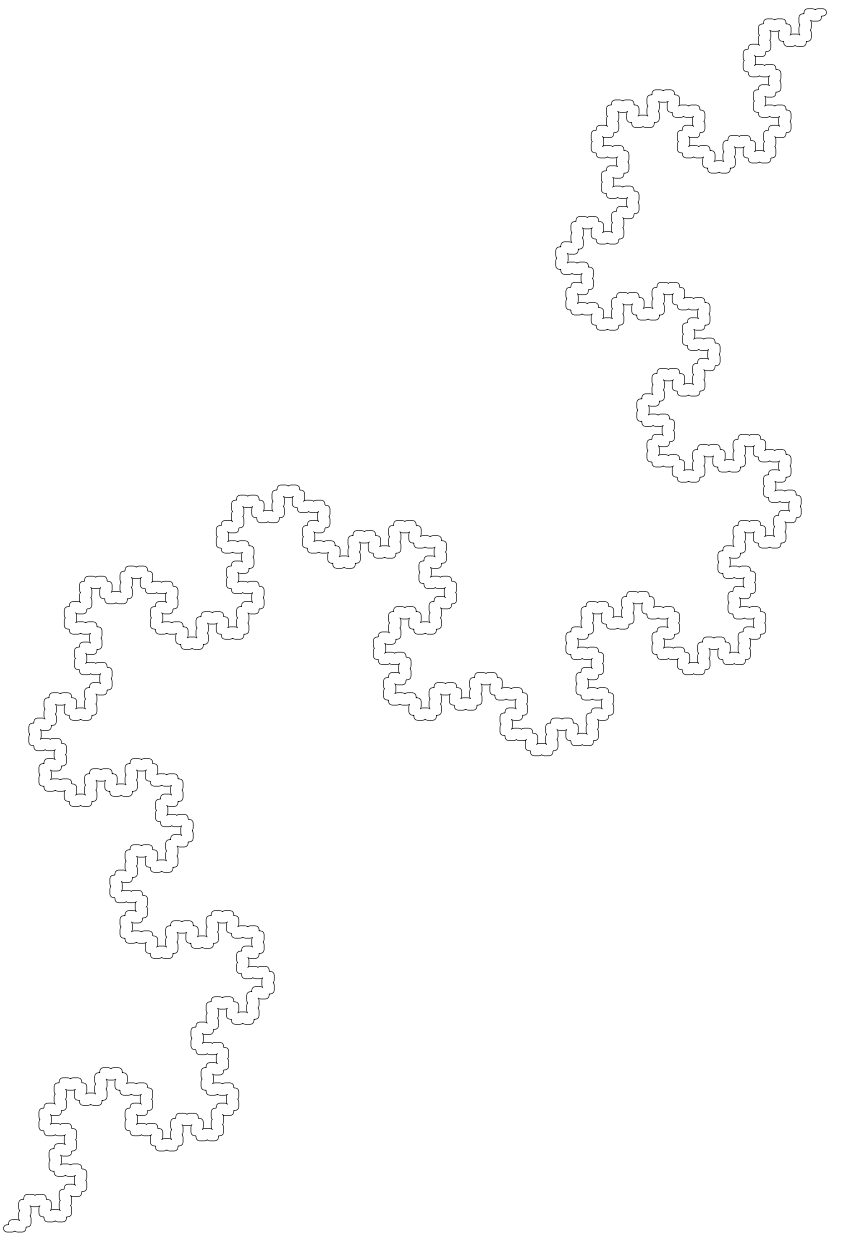
Mandalic Star

L-System Generator • 5 Iterations



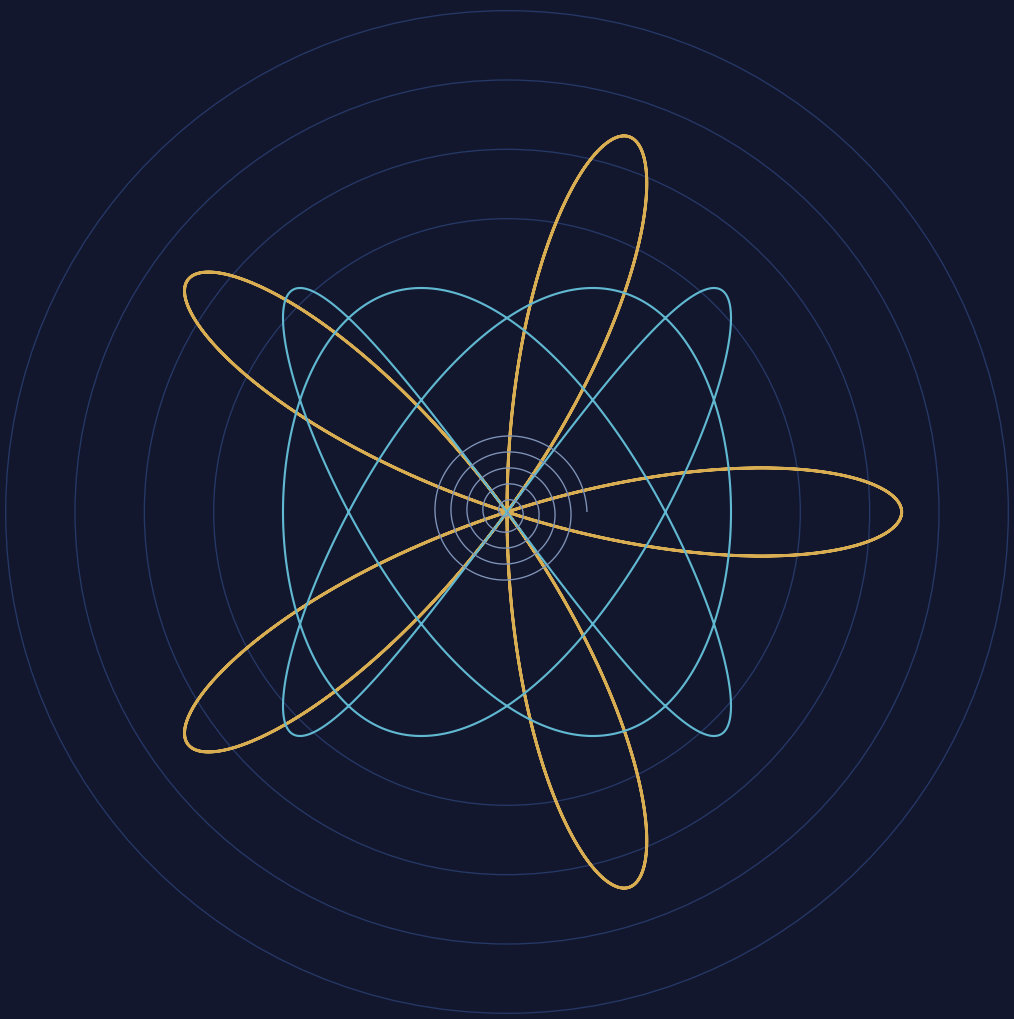
Geometric Flower

L-System Generator • 5 Iterations



Infinite Labyrinth

L-System Generator • 4 Iterations



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Thank you for coloring!

We hope you enjoyed every curve and equation.

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